

Econometrics of Commodity and Asset Pricing

Lecture 5

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- I. Estimation of Affine Asset-Pricing Models
- II. Pricing Credit and Liquidity Risks

Chapter I

ESTIMATION OF AFFINE ASSET-PRICING MODELS

– Agenda –

- I.1 State-Space Models. (Slide 4)
- I.2 Kalman-Filter-Based Approaches. (Slide 9)
- I.3 Inversion Technique. (Slide 20)
- I.4 Dealing with Unobserved Regimes: An Introduction. (Slide 25)
- I.5 Dealing with Unobserved Regimes. (Slide 37)
- I.6 The Persistence Problem. (Slide 46)
- ▶ Appendix I.A: Kitagawa-Hamilton filter.
- ▶ Appendix I.B: Kitagawa-Hamilton and inversion technique.

I.1 – STATE-SPACE MODEL

- Dynamic asset-pricing models we have seen in the previous lectures are **state-space models**. The dynamics of all variables, gathered in vector y_t (yields, equity returns, macroeconomic variables, survey-based variables) is accounted for by state variables (w_t).
- ▶ The equations defining the relationship between the other variables and the state variables are called **measurement equations** (Eq. I.1).
- ▶ The equations defining the dynamics of the state variables are called **transition equations** (Eq. I.2).
- ▶ If w_t is an affine process, the transition equations admit a VAR(1) representation (see Lecture 1).

Definition I.1 (Linear State-Space Model)

$$\begin{matrix} y_t \\ (m \times 1) \end{matrix} = A + Bw_t + \eta_t \quad \text{with } \eta_t \sim i.i.d. \mathcal{N}(0, \Omega) \quad (I.1)$$

$$\begin{matrix} w_t \\ (n \times 1) \end{matrix} = \mu + \Phi w_{t-1} + \Sigma^{\frac{1}{2}}(w_{t-1})\varepsilon_t, \quad (I.2)$$

where ε_t is a martingale difference sequence satisfying $\mathbb{V}ar_t(\varepsilon_{t+1}) = Id$. The components of η_t are **measurement errors**.

- ▶ There are two different situations:

- (a) all state variables (components of w_t) are observed (next slide);
- (b) some of these variables are latent.

What do we mean by “model estimation” in case (b)?

Three possible meanings:

- (b.i) We know the model parameters but we want to recover the latent factors (for instance to compute model-implied prices).
- (b.ii) We know neither the model parameters nor the latent variables, we want to estimate both of them.
- (b.iii) We know neither the model parameters nor the latent variables, we are just interested in the model parameters.

- **Case (a):** One can resort to standard estimation techniques (GMM, Maximum Likelihood) to estimate model parameters.

- Take the maximum likelihood case, we have:

$$\begin{aligned} f(y_t, w_t | \underline{y_{t-1}}, \underline{w_{t-1}}) &= f(y_t | \underline{y_{t-1}}, \underline{w_t}) \times \underbrace{f(w_t | \underline{y_{t-1}}, \underline{w_{t-1}})}_{= f(w_t | \underline{w_{t-1}})} \\ &= \mathbf{n}(y_t; A + Bw_t, \Omega) f(w_t | \underline{w_{t-1}}), \end{aligned}$$

where $\mathbf{n}(x; \mu, \Omega)$ denotes the evaluation, at vector x of the p.d.f. of the multivariate normal distribution $\mathbf{n}(\mu, \Omega)$, i.e., if x is a m -dimensional vector:

$$\mathbf{n}(x; \mu, \Omega) = \frac{1}{\sqrt{(2\pi)^m |\Omega|}} \exp \left(-\frac{1}{2} \{x - \mu\}' \Omega^{-1} \{x - \mu\} \right). \quad (1.3)$$

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- ▶ Once this conditional p.d.f. is known, the total likelihood is given by (conditional on y_0 and w_0):

$$\prod_{t=1}^T f(y_t, w_t | \underline{y_{t-1}}, \underline{w_{t-1}}).$$

- ▶ $f(w_t | \underline{w_{t-1}})$ depends on the process chosen for w_t . If complicated to compute, one can employ Pseudo ML (see here below).

Remark 1.1

- ▶ *In some cases, the computation of $f(w_t | \underline{w_{t-1}})$ may be complicated.*
- ▶ *For instance, the p.d.f. may involve an infinite sum, which is the case in the ARG case (see Lecture 1).*
- ▶ *When w_t is affine, the Pseudo Maximum Likelihood approach consists in replacing $f(w_t | \underline{w_{t-1}})$ by:*

$$n(w_t; \mu + \Phi w_{t-1}, \Sigma(w_t)).$$

where μ , Φ and $\Sigma(w_t)$ are given by the affine nature of the process.

□ **In case (b.iii):**

one can estimate the model by fitting sample moments computed using observed variables (prices, yields, returns) $\Rightarrow$ **Generalized Method of Moments (GMM)**.

- ▶ In the context of affine processes, conditional and unconditional mean and variance-covariance of the state vector w_t are available in closed form. Conditional cumulants are affine in the state vector (see Lecture 1).
- ▶ If the model-implied moments are not available in closed-form, one may have to resort to the **Simulated Method of Moments (SMM)** [Duffie and Singleton (1993) or Gouriéroux and Monfort (1996)].
- ▶ In cases (b.i) and (b.ii): one has to implement **filtering methods**.
- ▶ We focus on the latter context in the following slides.

I.2 – KALMAN-FILTER-BASED APPROACHES

- Let us start with the particular case of state-space model (Def. I.1) where ε_t is Gaussian and where $\Sigma^{\frac{1}{2}}$ does not depend on w_t , i.e. with a homoskedastic linear Gaussian state-space model.
- Let's denote by θ the vector of parameters that defines the model. For a given θ and a sequence of observations $\{y_1, \dots, y_T\}$:
 - the Kalman Filter (KF) computes the distribution of w_t given $\{y_1, \dots, y_t\}$ (see Def. I.2);
 - this distribution is Gaussian, and obtained by a recursive algorithm.
- Byproduct of the KF :
 - calculates the likelihood associated with θ ;
 - makes it possible to estimate θ by maximizing the likelihood (using a numerical algorithm);
 - hence, Kalman-filter techniques can address Objective (b.ii) (Slide 6).

Definition I.2 (Filtered versus smoothed estimates)

$$\text{Filtering:} \quad w_{t|t} = \mathbb{E}(w_t | \underline{y}_t) \quad \text{and} \quad P_{t|t} = \text{Var}(w_t | \underline{y}_t)$$

$$\text{Smoothing:} \quad w_{t|T} = \mathbb{E}(w_t | \underline{y}_T) \quad \text{and} \quad P_{t|T} = \text{Var}(w_t | \underline{y}_T)$$

Kalman Filter: Recursions

- When Σ does not depend on w_t (Eq. 1.2), the KF yields to the estimates of the w_t with the lowest possible mean squared errors.

Kalman filter algorithm

Iteration $t - 1$: $w_{t-1} | \underline{y}_{t-1} \sim \mathbf{n}(w_{t-1|t-1}, P_{t-1|t-1})$.

Iteration t – Prediction Step ($1 \leq t \leq T$):

- $w_{t|t-1} = \mu + \Phi w_{t-1,t-1}$;
- $P_{t|t-1} = \Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}'} + \Phi P_{t-1|t-1} \Phi'$;

$$w_t | \underline{y}_{t-1} \sim \mathbf{n}\left(\underbrace{\mu + \Phi w_{t-1,t-1}}_{w_{t|t-1}}, \underbrace{\Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}'} + \Phi P_{t-1|t-1} \Phi'}_{P_{t|t-1}}\right). \quad (1.4)$$

- $y_{t|t-1} = \mathbb{E}[y_t | \underline{y}_{t-1}] = A + B w_{t|t-1}$;
- $S_{t|t-1} = \mathbb{V}[y_t | \underline{y}_{t-1}] = B P_{t|t-1} B' + \Omega$;

Iteration t – Updating Step ($1 \leq t \leq T$):

$$f(w_t | \underline{y}_t) = f(y_t | w_t) \times f(w_t | \underline{y}_{t-1}) / f(y_t | \underline{y}_{t-1}), \quad (1.5)$$

which is Gaussian, i.e. $w_t | \underline{y}_t \sim \mathbf{n}(w_{t|t}, P_{t|t})$.

- + The conditional mean ($w_{t|t}$) linearly depends on y_t .

The updating formula is based on the following proposition:

Prop. 1.1 (Updates of Gaussian variables)

Assume that

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \sim \mathbf{n} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{bmatrix} \right).$$

The

$$y_2|y_1 \sim \mathbf{n}(\mathbb{E}[y_2|y_1], \mathbb{V}[y_2|y_1]) ,$$

where:

$$\mathbb{E}[y_2|y_1] = \mu_2 + \Omega_{21}\Omega_{11}^{-1}(y_1 - \mu_1) ,$$

$$\mathbb{V}[y_2|y_1] = \Omega_{22} - \Omega_{21}\Omega_{11}^{-1}\Omega_{12} .$$

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Kalman filter algorithm - Updating step ($1 \leq t \leq T$):

- $w_{t|t}$ and $P_{t|t}$ are given by:

$$w_{t|t} = w_{t|t-1} + K_t \lambda_t \quad \text{and} \quad P_{t|t} = (I - K_t B) P_{t|t-1},$$

where

$$\lambda_t = y_t - \mathbb{E}(y_t | \underline{y}_{t-1}) = y_t - A - B w_{t|t-1}$$

$$S_{t|t-1} = \text{Var}(y_t | \underline{y}_{t-1}) = B P_{t|t-1} B' + \Omega$$

$$K_t = P_{t|t-1} B' S_{t|t-1}^{-1} \quad (\text{Kalman gain}).$$

Kalman filter: Log-likelihood

- The log likelihood is (recursively) computed as follows:

$$\begin{aligned} \log \mathcal{L}(\theta; \underline{y}_T) &= \frac{mT}{2} \log(2\pi) \\ &\quad - \frac{1}{2} \sum_{t=1}^T \left(\log |S_{t|t-1}(\theta)| + \lambda_t'(\theta) S_{t|t-1}^{-1}(\theta) \lambda_t(\theta) \right). \end{aligned}$$

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Prop. 1.2 (Kalman filter and smoother (summary))

If $\varepsilon_t \sim \mathcal{N}(0, I)$ in the state-space defined in Def. 1.1, then we have (“filtering”):

$$w_t | y_1, \dots, y_t \sim \mathcal{N}(w_{t|t}, P_{t|t}),$$

where $w_{t|t}$ and $P_{t|t}$ result from the following recursive equations:

$$w_{t|t} = w_{t|t-1} + K_t \lambda_t$$

$$P_{t|t} = (I - K_t B) P_{t|t-1}$$

where (updating step)

$$\lambda_t = y_t - A - B w_{t|t-1} \quad (\text{forecast error})$$

$$S_{t|t-1} = \text{Var}(y_t | \underline{y_{t-1}}) = B P_{t|t-1} B' + \Omega$$

$$K_t = P_{t|t-1} B' S_{t|t-1}^{-1} \quad (\text{Kalman gain})$$

and where (prediction step)

$$w_{t|t-1} = \mu + \Phi w_{t-1|t-1}$$

$$P_{t|t-1} = \Sigma + \Phi P_{t-1|t-1} \Phi' \quad (\Sigma = \Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}'}).$$

The log likelihood is (recursively) computed as follows:

$$\log \mathcal{L}(\theta; \underline{y}_T) = \frac{mT}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left(\log |S_{t|t-1}(\theta)| + \lambda_t'(\theta) S_{t|t-1}^{-1}(\theta) \lambda_t(\theta) \right). \quad (1.6)$$

Moreover, we also have (“smoothing”):

$$w_t | y_1, \dots, y_T \sim \mathcal{N}(w_{t|T} | P_{t|T}),$$

where $w_{t|T}$ and $P_{t|T}$ result from the following recursive equations:

$$w_{t|T} = w_{t|t} + F_t(w_{t+1|T} - w_{t+1|t})$$

$$P_{t|T} = P_{t|t} + F_t(P_{t+1|T} - P_{t+1|t})F_t'$$

where

$$F_t = P_{t|t} \Phi_{t+1}' P_{t+1|t}^{-1}.$$

Remark 1.2 (Initialization)

- ▶ *Recursive procedure: initial conditions are required ("Iteration 0").*
- ▶ *There are different possibilities regarding the initialization of $(w_{0|0}, P_{0|0})$:*
 - A. *include the elements of $(w_{0|0}, P_{0|0})$ among the parameters to estimate;*
 - B. *set $(w_{0|0}, P_{0|0})$ to their unconditional values (marginal moments);*
 - C. *set $w_{0|0}$ to a prior value and take*
 - ▶ *an arbitrary large value for $P_{0|0}$ if the prior value is uncertain ($\sim$ "diffuse prior");*
 - ▶ *a small value for $P_{0|0}$ if we are confident in this prior value.*

Remark 1.3 (Missing data)

- ▶ *In many application, one do not observe all the entries of y_t at every date (because e.g. different frequencies, unreliable data for some period, short historical series for some variables).*
- ▶ *Easily addressed by Kalman filtering/smoothing: One simply has to change the size of y_t (and of λ_t , $S_{t|t-1}$, A and B) on each date.*
- ▶ *Of course, the greater the number of observations available to update the filter, the better the accuracy of the estimation.*
- ▶ *Log-likelihood computation (Eq. 1.6) still valid (adjust for the number of observed variables at each iteration; m then depends on time).*

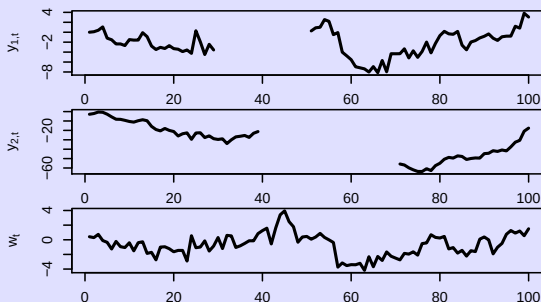
Remark 1.3 (Missing data (cont'd))

- Consider the following model

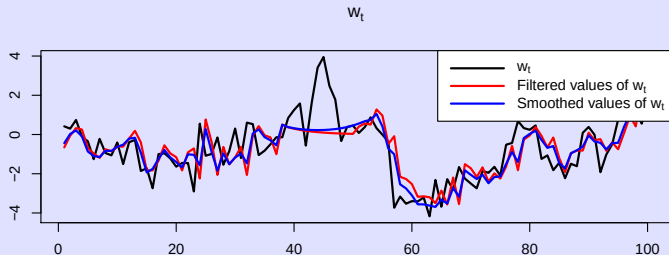
$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-2} \end{bmatrix} + \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} w_t + \eta_t \quad (1.7)$$

$$w_t = \phi w_{t-1} + \varepsilon_t. \quad (1.8)$$

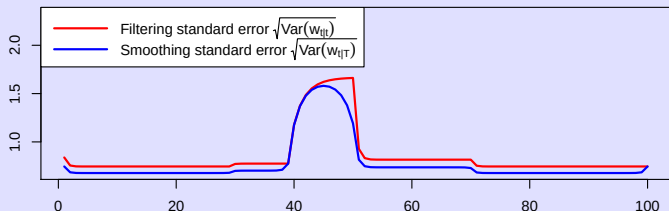
- ▶ We simulate a 100-period sample, we remove observations of $y_{1,t}$ (respectively of $y_{2,t}$) between periods $t = 30$ and $t = 50$ (resp. between periods $t = 40$ and $t = 70$), and try to recover the states w_t .



Remark 1.3 (Missing data (cont'd))



Conditional standard deviation of filtered/smoothed estimates



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Remark 1.4 (About non-constant conditional matrix Σ)

- ▶ Proposition 1.2 is valid when ε_t is Gaussian and when $\Sigma^{\frac{1}{2}}$ does not depend on w_t .
- ▶ How to proceed in the general case?
- ▶ In the general case (but when w_t is an affine process), we have that $\Sigma(w_t) \equiv \text{Var}(w_{t+1}|\underline{w}_t)$ is affine in w_t .
- ▶ In order to deal with this, the Kalman filter algorithm can be modified.
- ▶ Specifically, in the prediction step (see Prop. 1.2), $P_{t|t-1}$ can be computed as:

$$P_{t|t-1} = \Sigma(w_{t-1|t-1}) + \Phi P_{t-1|t-1} \Phi',$$

i.e. we replace $\Sigma(w_{t-1})$ by $\Sigma(w_{t-1|t-1})$.

- ▶ Though the approach is then not necessarily optimal, it shows good empirical properties [Duan and Simonato (1999) or de Jong (2000)].
- ▶ In order to test for the validity of the approach in a specific context, one can resort to Monte-Carlo simulations [see e.g. Monfort, Pegoraro, Renne and Roussellet (2017)].

Remark 1.5 (Non-linear models)

- ▶ If w_t follows an affine process, it admits the VAR dynamics: $\Rightarrow$ linear transition equation.
- ▶ However, measurement equations may be non-linear functions of w_t .
- ▶ This is in particular the case if observed variables include Swaps rates, CDS rates or prices of tranche products.
- ▶ In this case, one can for instance resort to the Extended Kalman Filter (linearizing the measurement equations, see supplementary material) or the Quadratic Kalman Filter (Monfort, Renne and Roussellet, 2015, second-order expansion of the measurement equations).

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I.3 – INVERSION TECHNIQUE

- ▶ The **inversion technique** has been introduced by **Chen and Scott (1993)**.
- ▶ It is used e.g. by **Ang and Piazzesi (2003)** and **Liu, Longstaff and Mandell (2006)**. See **Hamilton and Wu (2012)** for associated identification and estimation issues concerning GATSMs.
- ▶ Contrary to Kalman-type approaches, this approach is not recursive. Hence can be faster (especially for long sample – in time dimension).
- ▶ Recall that y_t and w_t are respectively of dimension m and n (see Eqs. **I.1** and **I.2** in Def. **I.1**).

Assumption I.1 (Perfectly-modelled variables)

- ▶ *Assumption: n components of the m -dimensional vector y_t (with $n \leq m$) are perfectly modelled (no measurement errors in associated measurement equations).*
- ▶ *Without loss of generality, these perfectly-modelled variables are the first n components of y_t , that is:*

$$y_t = \begin{pmatrix} \underbrace{y_{1,t}}_{(n \times 1)} \\ \underbrace{y_{2,t}}_{(m-n) \times 1} \end{pmatrix}.$$

- Under Assumption I.1, the measurement equation (Eq. I.1) becomes:

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} w_t + \begin{bmatrix} 0 \\ \eta_{2,t} \end{bmatrix},$$

where $\eta_{2,t} \sim \mathcal{N}(0, \Omega_2)$ (say). This notably implies

$$w_t = w(y_{1,t}) = B_1^{-1}(y_{1,t} - A_1). \quad (\text{I.9})$$

- Under this assumption and if the conditional distribution of w_t is available in closed form, then the (exact) likelihood of the model can then be computed.
- Proposition I.3 shows (here below) that the computation of $f_{Y_t|Y_{t-1}}(y_t; \underline{y}_{t-1})$ involves three terms:
- The **first term** stems from the conditional distribution $w_t | \underline{w}_{t-1}$.
 - The **second term** is associated with the measurement errors pertaining to $y_{2,t}$, that are the components of $\eta_{2,t}$.
 - The **third term** is the determinant of the Jacobian matrix associated with the linear transformation between w_t and $y_{1,t}$ (Eq. I.9), that is $|B_1|$.
- Once one knows how to compute $f_{Y_t|Y_{t-1}}(y_t; \underline{y}_{t-1})$, the total likelihood is easily obtained since:

$$f_{Y_1, \dots, Y_T}(y_1, y_2, \dots, y_T) = f_{Y_1}(y_1) \prod_{t=2}^T f_{Y_t|Y_{t-1}}(y_t; \underline{y}_{t-1}).$$

- In order to properly present Proposition I.3, we highlight the following results. Given I.9 and the historical dynamics of (w_t) , we can write:

$$B_1^{-1}(y_{1,t} - A_1) = \mu + \Phi \left[B_1^{-1}(y_{1,t-1} - A_1) \right] + \Sigma \varepsilon_t ,$$

thus implying:

$$\begin{aligned} y_{1,t} &= \left[B_1 \mu + A_1 - B_1 \Phi B_1^{-1} A_1 \right] + \left(B_1 \Phi B_1^{-1} \right) y_{1,t-1} + B_1 \Sigma \varepsilon_t , \\ &= \mu_y + \Phi_y y_{1,t-1} + \Sigma_y \varepsilon_t . \end{aligned} \tag{I.10}$$

In other words, $y_{1,t}$ is a (Markovian) Gaussian VAR(1) process with $f_1(y_{1,t} | y_{1,t-1}) = \mathbf{n}(y_{1,t}; \mu_y + \Phi_y y_{1,t-1}, \Sigma_y \Sigma_y')$. Moreover:

$$\begin{aligned} y_{2,t} &= A_2 + B_2 w_t + \eta_{2,t} = A_2 + B_2 B_1^{-1}(y_{1,t} - A_1) + \eta_{2,t} , \\ &= (A_2 - B_2 B_1^{-1} A_1) + B_2 B_1^{-1} y_{1,t} + \eta_{2,t} , \\ &= \tilde{A}_2 + \tilde{B}_2 y_{1,t} + \eta_{2,t} , \end{aligned} \tag{I.11}$$

that is

$$f_2(y_{2,t} | y_{1,t}) = \mathbf{n}(y_{2,t}; A_2 + B_2 w(y_{1,t}), \Omega_2) = \mathbf{n}(y_{2,t}; \tilde{A}_2 + \tilde{B}_2 y_{1,t}, \Omega_2).$$

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- Let us remember the following : if U and V are two random variables such that $V = g(U)$, where g is a bijective, invertible and differentiable function, then we have :

$$f_V(v) = \left| \frac{\partial g^{-1}(v)}{\partial v'} \right| f_U(g^{-1}(v)).$$

- We want to determine $f_1(y_{1,t} | y_{1,t-1})$ and we know that:

$$\begin{aligned} y_{1,t} &= A_1 + B_1 w_t = g(w_t), \text{ with} \\ f_w(w_t | w_{t-1}) &= \mathbf{n}(w_t; \mu + \Phi w_{t-1}, \Sigma \Sigma'). \end{aligned}$$

- We thus obtain

$$f_1(y_{1,t} | y_{1,t-1}) = |B_1|^{-1} f_w(w(y_{1,t}) | w(y_{1,t-1})), \text{ being}$$

$$w(y_{1,t}) = g^{-1}(y_{1,t}) = B_1^{-1}(y_{1,t} - A_1) \text{ and } \left| \frac{\partial g^{-1}(y_{1,t})}{\partial y'_{1,t}} \right| = |B_1|^{-1},$$

and the joint conditional p.d.f. is:

$$\begin{aligned} & f_{Y_t | Y_{t-1}}(y_t; y_{t-1}) \\ &= \mathbf{n}(y_{1,t}; \mu_y + \Phi_y y_{1,t-1}, \Sigma_y \Sigma'_y) \mathbf{n}(y_{2,t}; \tilde{A}_2 + \tilde{B}_2 y_{1,t}, \Omega_2) \\ &= |B_1|^{-1} f_w(w(y_{1,t}) | w(y_{1,t-1})) \mathbf{n}(y_{2,t}; A_2 + B_2 w(y_{1,t}), \Omega_2). \end{aligned}$$

In the context of a linear state-space model as defined in Def. 1.1, under Assumption 1.1, and if w_t is a Markovian process, we have:

$$f_{Y_t|\underline{Y_{t-1}}}(y_t; \underline{y_{t-1}}) = f_{w_t|w_{t-1}}(w(y_{1,t}); w(y_{t-1})) \times \mathbf{n}(y_{2,t}; A_2 + B_2 w(y_{1,t}), \Omega_2) \times |B_1|^{-1}.$$

where $w(y_{1,t}) = B_1^{-1}(y_{1,t} - A_1)$ and where $\mathbf{n}$ denotes the multivariate normal p.d.f. (Eq. I.3).

- The likelihood function is thus given by:

$$\begin{aligned} & \mathcal{L}(\theta, Y) \\ = & f_{Y_1}(y_1) \prod_{t=2}^T |B_1|^{-1} f_w(w(y_{1,t}) | w(y_{1,t-1})) \mathbf{n}(y_{2,t}; A_2 + B_2 w(y_{1,t}), \Omega_2), \end{aligned}$$

and we do not have Internal Consistency Conditions (ICC). The Chen-Scott procedure then maximize $\mathcal{L}(\theta, Y)$ w.r.t. θ ; this numerical search can be very difficult and reaching the true global MLE complicated (see [Hamilton and Wu \(2012\)](#) for further developments).

- Observe that $y_t = (y'_{1,t}, y'_{2,t})'$ is a (restricted) Gaussian VAR(1) process in block-recursive form.

I.4 – DEALING WITH UNOBSERVED REGIMES: AN INTRODUCTION

- The (adjusted) Kalman is not suited to the case where some of the components of w_t are valued in a discrete set. This is typically the case if w_t is of the form:

$$w_t = \begin{pmatrix} y_t \\ z_t \end{pmatrix},$$

where z_t is valued in $\{e_1, \dots, e_J\}$, e_j being the j^{th} column of Id_J .

- Let us start introducing the problem by means of the following regression model with structural breaks in the parameters:

$$y_t = x_t (\beta' z_t) + e_t, \quad e_t \sim N(0, \sigma^2 z_t),$$

$$\beta = (\beta_1, \beta_2)', \quad \sigma^2 = (\sigma_1^2, \sigma_2^2)',$$

z_t is valued in $\{e_1, e_2\}$, x_t is an exogenous variable.

- If z_t is known a priori for any date $t = 1, \dots, T$, that is the dates of the regime switching (or structural breaks) are known a priori, the above is nothing more than a dummy variable.

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- In this case the log-likelihood function is given by:

$$\ln L = \sum_{t=1}^T \ln(f(y_t | z_t)),$$
$$f(y_t | z_t) = \frac{1}{\sqrt{2\pi(\sigma^2 z_t)}} \exp\left(-\frac{(y_t - x_t(\beta' z_t))^2}{2(\sigma^2 z_t)}\right),$$

and it can be maximized w.r.t. $(\beta_1, \beta_2, \sigma_1^2, \sigma_2^2)'$.

- A major problem arises when z_t is not observed (at any date t). In this case, as a rule of thumb, we use the following two steps to determine the log-likelihood function:

Step 1. First, consider the joint density of (y_t, z_t) which is given by the product of conditional and marginal densities:

$$f(y_t, z_t | \underline{y_{t-1}}) = f(y_t | z_t, \underline{y_{t-1}}) f(z_t | \underline{y_{t-1}}).$$

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Step 2. Then, to obtain the density of the observable y_t , given its own past only, we have to integrate z_t out of the above joint density by summing over all possible values of z_t :

$$\begin{aligned} f(y_t | \underline{y}_{t-1}) &= \sum_{j=1}^2 f(y_t, z_t = e_j | \underline{y}_{t-1}) \\ &= \sum_{j=1}^2 f(y_t | z_t = e_j, \underline{y}_{t-1}) f(z_t = e_j | \underline{y}_{t-1}) \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(y_t - x_t\beta_1)^2}{2\sigma_1^2}\right) \times f(z_t = e_1 | \underline{y}_{t-1}), \\ &\quad + \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(y_t - x_t\beta_2)^2}{2\sigma_2^2}\right) \times f(z_t = e_2 | \underline{y}_{t-1}). \end{aligned}$$

- The log-likelihood function then is given by:

$$\ln L = \sum_{t=1}^T \ln \left[\sum_{j=1}^2 f(y_t | z_t = e_j, \underline{y}_{t-1}) f(z_t = e_j | \underline{y}_{t-1}) \right].$$

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- To calculate the log-likelihood function we need to calculate the weighting factors $f(z_t = e_1 | y_{t-1})$ and $f(z_t = e_2 | y_{t-1})$.
- Nevertheless, without a priori assumptions about the stochastic behavior of the z_t variable, this will not be possible.
- In the following slides we will consider different assumptions about the evolution of z_t .
- Case 1 - Independent switching : when z_t evolves independently of its own past values, we can specify probabilities very simply as:

$$\mathbb{P}[z_t = e_1] = p = \frac{\exp(p_1)}{1 + \exp(p_1)}$$

$$\mathbb{P}[z_t = e_2] = 1 - p = 1 - \frac{\exp(p_1)}{1 + \exp(p_1)},$$

where p_1 is an unconstrained parameter. Furthermore, if the stochastic behavior of z_t is not dependent upon any other exogenous or predetermined variable, we have $f(z_t = e_j | y_{t-1}) = \mathbb{P}[z_t = e_j]$ and the log-likelihood function can be maximized w.r.t. $(\beta_1, \beta_2, \sigma_1^2, \sigma_2^2, p_1)'$.

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- Case 2 - Markov switching: the evolution of z_t may depend upon $z_{t-1}, z_{t-2}, \dots, z_{t-r}$, in which case the process is named as an r -th order Markov switching process.
- Here we consider the simplest case of a two-state, first-order Markov-switching process for z_t , with the following transition probabilities:

$$\mathbb{P}[z_t = e_1 \mid z_{t-1} = e_1] = p = \frac{\exp(p_1)}{1 + \exp(p_1)}$$

$$\mathbb{P}[z_t = e_2 \mid z_{t-1} = e_2] = q = \frac{\exp(q_1)}{1 + \exp(q_1)}.$$

- Dealing with the problem of unobserved z_t in the above model is exactly the same as before.
- Nevertheless, determining the weighting terms $f(z_t = e_j \mid \underline{y_{t-1}})$ is not as straightforward as before.

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- We adopt the following filter to calculate the weighting terms:

Step 1. Given $f(z_{t-1} = e_j | \underline{y_{t-1}})$, $j = 1, 2$, at the beginning of time t (or at the t -th iteration of the filter), the weighting terms $f(z_t = e_j | \underline{y_{t-1}})$, $j = 1, 2$, are calculated as:

$$\begin{aligned} f(z_t = e_j | \underline{y_{t-1}}) &= \sum_{i=1}^2 f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}}) \\ &= \sum_{i=1}^2 \mathbb{P}[z_t = e_j | z_{t-1} = e_i] f(z_{t-1} = e_i | \underline{y_{t-1}}), \end{aligned}$$

where $\mathbb{P}[z_t = e_j | z_{t-1} = e_i]$ are the transition probabilities.

Step 2. Once y_t is observed at the end of time t (or at the end of the t -th iteration of the filter), we can update the probability term in the following way:

$$\begin{aligned} f(z_t = e_j | \underline{y_t}) &= f(z_t = e_j | y_t, \underline{y_{t-1}}) = \frac{f(z_t=e_j, y_t | \underline{y_{t-1}})}{f(y_t | \underline{y_{t-1}})} \\ &= \frac{f(y_t | z_t=e_j, \underline{y_{t-1}}) f(z_t=e_j | \underline{y_{t-1}})}{\sum_{j=1}^2 f(y_t | z_t=e_j, \underline{y_{t-1}}) f(z_t=e_j | \underline{y_{t-1}})}. \end{aligned}$$

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- The above two steps may be iterated to get $f(z_t = e_j | \underline{y_{t-1}})$ for any $t = 1, \dots, T$.
- To start the above filter at time $t = 1$, however, we need $f(z_t | y_0)$. We can employ the steady-state or unconditional probabilities of z_t :

$$\pi_1 = f(z_t = e_1 | y_0) = \frac{1-p}{2-p-q},$$

$$\pi_2 = f(z_t = e_2 | y_0) = \frac{1-q}{2-p-q}.$$

- It is also clear that the log-likelihood function has $(\beta_1, \beta_2, \sigma_1^2, \sigma_2^2, p_1, q_1)'$ as unknown parameters.
- Case 3 - Serially correlated data and Markov switching: let us consider the following Gaussian AR(1) process:

$$(y_t - \mu' z_t) = \phi_1 (y_{t-1} - \mu' z_{t-1}) + e_t, \quad e_t \sim i.i.d.N(0, \sigma^2 z_t).$$

where z_t is a J -state homogeneous Markov chain.

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- In writing the density of y_t given $\underline{y_{t-1}}$, we need z_t and z_{t-1} , which are unobserved. To solve the problem, we proceed as before, except that we now consider the joint density of y_t , z_t and z_{t-1} :

Step 1. Derive the joint density of y_t , z_t and z_{t-1} , conditional on $\underline{y_{t-1}}$:

$$\begin{aligned}f(y_t, z_t, z_{t-1} | \underline{y_{t-1}}) &= f(y_t | z_t, z_{t-1}, \underline{y_{t-1}}) f(z_t, z_{t-1} | \underline{y_{t-1}}), \\f(y_t | z_t, z_{t-1}, \underline{y_{t-1}}) &= \frac{1}{\sqrt{2\pi(\sigma^2 z_t')}} \exp\left(-\frac{[(y_t - \mu' z_t) - \phi_1(y_{t-1} - \mu' z_{t-1})]^2}{2(\sigma^2 z_t')}\right).\end{aligned}$$

Step 2. In order to calculate $f(y_t | \underline{y_{t-1}})$, we need to integrate z_t and z_{t-1} out of the joint density by summing the joint density over all possible values of z_t and z_{t-1} :

$$\begin{aligned}f(y_t | \underline{y_{t-1}}) &= \sum_{j=1}^J \sum_{i=1}^J f(y_t, z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}}) \\&= \sum_{j=1}^J \sum_{i=1}^J f(y_t | z_t = e_j, z_{t-1} = e_i, \underline{y_{t-1}}) f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}}).\end{aligned}$$

where the density $f(y_t | \underline{y_{t-1}})$ is a weighted average of J^2 conditional densities with weights $f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}})$.

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- Then, the log-likelihood function is given by:

$$\ln L = \sum_{t=1}^T \ln \left[\sum_{j=1}^J \sum_{i=1}^J f(y_t | z_t = e_j, z_{t-1} = e_i, \underline{y_{t-1}}) f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}}) \right].$$

- To complete the above procedure, we still need to deal with the problem of calculating $f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}})$.
- The following two (filtering) steps enable us to calculate those terms.

Step 1. Given $f(z_{t-1} = e_i | \underline{y_{t-1}})$, $i = 1, \dots, J$, at the beginning of time t (or at the t -th iteration of the filter), the weighting terms $f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}})$, $i, j = 1, \dots, J$, are calculated as:

$$f(z_t = e_j, z_{t-1} = e_i | \underline{y_{t-1}}) = \mathbb{P}[z_t = e_j | z_{t-1} = e_i] f(z_{t-1} = e_i | \underline{y_{t-1}})$$

where $\mathbb{P}[z_t = e_j | z_{t-1} = e_i]$ are the transition probabilities.

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Step 2. Once y_t is observed at the end of period t (or at the end of the t -th iteration), we can update the probability terms in the following way:

$$\begin{aligned} & f(z_t = e_j, z_{t-1} = e_i | \underline{y}_t) \\ = & f(z_t = e_j, z_{t-1} = e_i | \underline{y}_{t-1}, y_t) \\ = & \frac{f(z_t=e_j, z_{t-1}=e_i, y_t | \underline{y}_{t-1})}{f(y_t | \underline{y}_{t-1})} \\ = & \frac{f(y_t | z_t=e_j, z_{t-1}=e_i, \underline{y}_{t-1}) f(z_t=e_j, z_{t-1}=e_i | \underline{y}_{t-1})}{\sum_{j=1}^J \sum_{i=1}^J f(y_t | z_t=e_j, z_{t-1}=e_i, \underline{y}_{t-1}) f(z_t=e_j, z_{t-1}=e_i | \underline{y}_{t-1})} \end{aligned}$$

with the date- t filtered probability given by:

$$f(z_t = e_j | \underline{y}_t) = \sum_{i=1}^J f(z_t = e_j, z_{t-1} = e_i | \underline{y}_t).$$

- Iterating the above two steps for $t = 1, \dots, T$ provides us with the appropriate weighting terms.

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- To start the above filter at time $t = 1$, we can once again employ the steady-state or unconditional probabilities. For a two-state, first-order Markov switching, the steady-state probabilities are given by:

$$\pi_1 = f(z_t = e_1 | y_0) = \frac{1-p_{22}}{2-p_{22}-p_{11}},$$

$$\pi_2 = f(z_t = e_2 | y_0) = \frac{1-p_{11}}{2-p_{22}-p_{11}}.$$

- Let us see now the general derivation of those steady-state probabilities for a first-order J -state Markov switching process with transition probability matrix $\Pi = (p_{i,j})$.
- Given that the sum of the elements in each column of Π is equal to one, we have $\mathbf{1}_J' \Pi = \mathbf{1}_J'$, where $\mathbf{1}_J$ is a J -dimensional unitary vector.
- If we denote by π_t the date- t vector J steady-state probabilities, we have $\pi_t = [\mathbb{P}(z_t = e_1), \dots, \mathbb{P}(z_t = e_J)]' = [\pi_{1,t}, \dots, \pi_{J,t}]'$ and $\mathbf{1}_J' \pi_t = 1$.

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- Then, according to the definition of steady state probabilities, we have that $\pi_{t+1} = \Pi \pi_t$ and $\pi_{t+1} = \pi_t$, and thus:

$$\pi_t = \Pi \pi_t \Rightarrow (I_J - \Pi) \pi_t = \mathbf{0}_J,$$

where $\mathbf{0}_J$ is a J -dimensional vector of zeros. Combining this condition with $\mathbf{1}_J' \pi_t = 1$ we obtain:

$$\begin{bmatrix} I_J - \Pi \\ \mathbf{1}_J' \end{bmatrix} \pi_t = \begin{bmatrix} \mathbf{0}_J \\ 1 \end{bmatrix},$$

or

$$A \pi_t = \begin{bmatrix} \mathbf{0}_J \\ 1 \end{bmatrix}.$$

- If we multiply both sides of the above equation by $(A'A)^{-1} A'$. Then,

$$\pi_t = (A'A)^{-1} A' \begin{bmatrix} \mathbf{0}_J \\ 1 \end{bmatrix}.$$

That is, the matrix of steady-state probabilities, π_t , is the last column of the matrix $(A'A)^{-1} A'$.

I.5 – DEALING WITH UNOBSERVED REGIMES

- As previously mentioned, the (adjusted) Kalman and inversion techniques are not suited to the case where some of the components of w_t are valued in a discrete set, and this is the case when:

$$w_t = \begin{pmatrix} z_t \\ x_t \end{pmatrix},$$

where z_t is valued in $\{e_1, \dots, e_J\}$, e_j being the j^{th} column of Id_J .

- ▶ We assume again that z_t is an exogenous and homogenous Markov chain whose dynamics is defined by the $\pi(e_i, e_j)$'s that are such that:

$$\pi(e_i, e_j) = \mathbb{P}(z_{t+1} = e_j | z_t = e_i). \quad (I.12)$$

- We denote by Π the matrix of transition probabilities, i.e. the (i, j) component of Π is $\pi(e_i, e_j)$. Assume further that we have:

$$x_t = m(z_t, x_{t-1}) + \varepsilon_t, \quad (I.13)$$

where $\mathbb{E}(\varepsilon_t | \underline{z_t, x_{t-1}}) = 0$. The conditional distribution of ε_t w.r.t. (z_t, x_{t-1}) is denoted by $f_\varepsilon(\cdot; z_t, x_{t-1})$.

Assumption 1.2 (Measurement equations)

The measurement equation is of the form:

$$y_t = Az_t + Bx_t + \eta_t, \quad \text{with } \eta_t \sim i.i.d. \mathbf{n}(0, \Omega). \quad (1.14)$$

Example 1.1 (Regime-Switching Gaussian VAR)

- We know that if

$$x_t = \mu z_t + \Phi x_{t-1} + \varepsilon_t, \quad (1.15)$$

where $\varepsilon_t | x_t, z_t \sim \mathcal{N}(0, \Sigma(z_t))$ and if z_t is an exogenous independent Markov chain, then $w_t = (x_t', z_t')'$ is affine.

- This context is consistent with the situation presented on Slide 37. In particular, using the notations of Eq. (1.13), we have:

$$m(z_t, x_{t-1}) = \mu z_t + \Phi x_{t-1}.$$

- If r_t and the s.d.f. are respectively affine and exponential in w_t , then, in particular, yields are also affine in w_t , i.e. of the form $R(t, h) = A_h' z_t + B_h' x_t$.
- Therefore, if the components of y_t are yields of different maturities, the measurement equations are consistent with Assumption 1.2.

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- How to estimate such a model when the regimes z_t are unobservable?
Two distinct situations.

• **Case 1.** The x_t factors are observable.

- The probabilities of being in the different regimes on each date can be estimated by employing the Kitagawa-Hamilton filter.
- The Kitagawa-Hamilton filter (Appendix I.A) can be employed, with $F_t = (y'_t, x'_t)'$ and:

$$\begin{aligned} f(F_t | z_t = e_j, \underline{F_{t-1}}) &= f(y_t | x_t, z_t = e_j, \underline{F_{t-1}}) f(x_t | z_t = e_j, \underline{F_{t-1}}) \\ &= \mathbf{n}(y_t; A z_t + B x_t, \Omega) \times \\ &\quad f_\varepsilon(x_t - m(z_t, x_{t-1}); z_t, x_{t-1}), \end{aligned}$$

where, as in Slide 24, $\mathbf{n}(u; \mu, \Omega)$ denotes the evaluation, at vector u of the p.d.f. of the multivariate normal distribution $\mathcal{N}(\mu, \Omega)$.

- A by-product of the Kitagawa-Hamilton filter is the likelihood function associated with the dataset $\Rightarrow$ The model parameterization can be estimated by MLE.

- Case 2. The x_t factors are not observable.
 - ▶ Two sub-cases:
 - (2.i) The components of y_t are not perfectly modelled (i.e. $\Omega \neq 0$, where Ω defined in Eq. I.14).
 - (2.ii) n_x components of y_t are perfectly modelled (where n_x is the dimension of x_t).
- In Case (2.i), one has to resort to filters dealing with two types of uncertainty [with hidden discrete values (z_t) and continuously distributed latent variables (x_t)].
- In particular Kim (1994)'s filter can be employed when the state-space model is of the form (I.14)-(I.15) [Monfort and Renne (2014)].
- In Case (2.ii), one can resort to an inversion technique, complemented with the Kitagawa-Hamilton filter, to estimate the model (see following box and Dai, Singleton and Yang (2007)).

Mixed use of Kitagawa-Hamilton and inversion techniques

- Without loss of generality, assume that the n_x first components of y_t are observed without error, i.e. Assumption 1.2 becomes

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} A_1 z_t \\ A_2 z_t \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} x_t + \begin{bmatrix} 0 \\ \eta_{2,t} \end{bmatrix},$$

where $\varepsilon_2 \sim \mathcal{N}(0, \Omega_2)$.

- Since $y_{1,t} = A_1 z_t + B_1 x_t$, we then have:

$$x_t \equiv x(y_t, z_t) = B_1^{-1}(y_{1,t} - A_1 z_t). \quad (1.16)$$

- We can therefore write:

$$B_1^{-1}(y_{1,t} - A_1 z_t) = \mu z_t + \Phi [B_1^{-1}(y_{1,t-1} - A_1 z_{t-1})] + \varepsilon_t, \quad (1.17)$$

thus implying a Gaussian regime-switching VAR dynamics for $y_{1,t}$.

Mixed use of Kitagawa-Hamilton and inversion techniques (cont'd)

- In order to employ the Kitagawa-Hamilton filter (Appendix I.A), one need to define the extended Markov chain:

$$\mathcal{Z}_t = z_{t-1} \otimes z_t,$$

whose matrix of transition probabilities is detailed in Appendix I.B.

- Note that we have:

$$\begin{cases} z_t & \equiv & z(\mathcal{Z}_t) & = & (\mathbf{1}' \otimes Id_n) \mathcal{Z}_t \\ z_{t-1} & \equiv & z_{-1}(\mathcal{Z}_t) & = & (Id_n \otimes \mathbf{1}') \mathcal{Z}_t. \end{cases}$$

- The Kitagawa-Hamilton filter (Appendix I.A) can then be employed, with $F_t = y_t$ and:

$$f(y_t | \mathcal{Z}_t, \underline{y_{t-1}}) = |B_1^{-1}| \times f\left(\begin{bmatrix} x(y_t, z(\mathcal{Z}_t)) \\ y_{2,t} \end{bmatrix} | \mathcal{Z}_t, \underline{y_{t-1}}\right), \quad (I.18)$$

where the computation of the last term is detailed in Appendix I.B.

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Example 1.2 (Interest-rate model with monetary policy-related regimes)

- ▶ *Renne (2017): model where regimes have monetary-policy interpretations.*
- ▶ *Short-term rate = (daily) EONIA, euro-area overnight interbank rate:*

$$r_t = \underbrace{\bar{r}_t}_{\text{Target}} + \underbrace{x_t}_{\text{EONIA spread}}$$

- ▶ *Target rate $\bar{r}_t$ has a step-like path $\bar{r}_t = \Delta' z_{r,t}$, where*
 - ▶ $\Delta = [0 \quad 0.25 \quad 0.50 \quad \dots \quad \bar{r}_{\max}]'$ and
 - ▶ $z_{r,t} = [0 \quad \dots \quad 0 \quad 1 \quad 0 \quad \dots \quad 0]'$
- ▶ *EONIA spread (x_t) persistent and mean-reverting fluctuations (AR process).*
- ▶ $z_t = z_{r,t} \otimes z_{m,t}$ where $z_{m,t}$ is the monetary-policy regime:
 - ▶ **E**asing ($z_{m,t} = [1 \quad 0 \quad 0]'$),
 - ▶ **S**tatus Quo ($z_{m,t} = [0 \quad 1 \quad 0]'$),
 - ▶ **T**ightening ($z_{m,t} = [0 \quad 0 \quad 1]'$).

▶ Web-interface illustrating $\bar{r}_t$'s dynamics

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Example 1.2 (Interest-rate model with monetary policy-related regimes (cont'd))

- ▶ $z_{r,t}$ is observed, but not $z_{m,t}$.
- ▶ In the model, according to Example 1.1, we have:

$$R(t, h) = A_h' z_t + B_h x_t.$$

- ▶ Denote by $\mathcal{A}_h$ the $(3 \times n_r)$ matrix such that $A_h = \text{vec}(\mathcal{A}_h)$. We then have $A_h' z_t = (\mathcal{A}_h z_{r,t})' z_{m,t}$ and, therefore:

$$R(t, h) = A_{t,h}' z_t + B_h x_t, \quad \text{where } A_{t,h} = \mathcal{A}_h z_{r,t}.$$

- ▶ The model is estimated by a mixe use of Kitagawa-Hamilton and inversion techniques, assuming that a linear combination of yields is modelled without errors (giving $x_t = x(y_t, z_{m,t}, z_{r,t})$).

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Example 1.2 (Interest-rate model with monetary policy-related regimes (cont'd))

Figure 1: *Estimation outputs from Renne (2017)*

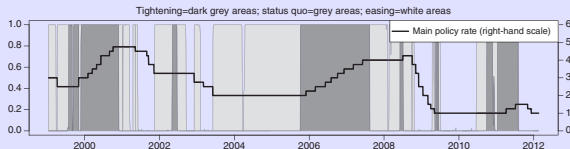


Figure 3: Regimes' estimates.

This figure displays the (smoothed) probabilities of being in the different monetary-policy regimes (left-hand scale). The main policy rate is also shown (right-hand scale).

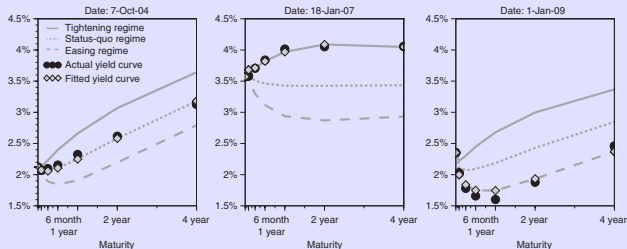


Figure 4: Fitted yield curves and influence of monetary-policy regimes.

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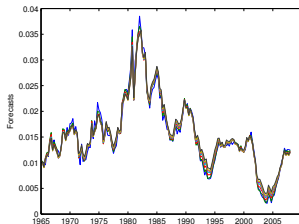
- ▶ Interest rates are particularly persistent variables.
- ▶ Since affine models eventually lead to linear relationships between state variables and interest rates, some state variables are also necessarily highly **persistent**.
- ▶ In small sample, maximum-likelihood estimates of the model parameters are likely to suffer from a downward bias [Bauer, Rudebusch and Wu (2012) or Jardet, Monfort and Pegoraro (2013)].
- ▶ This relates to what has been called **the persistence problem**.

I.6 – THE PERSISTENCE PROBLEM

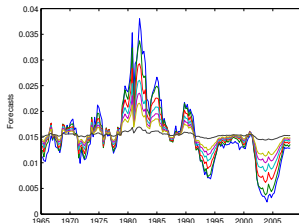
As discussed by [Jardet, Monfort and Pegoraro \(2013\)](#), the "**persistence problem**" can be decomposed into two subproblems:

- a) the "**discontinuity problem**" implied by the huge difference between [predictions based on models](#) taking into account unit [roots/cointegration](#) constraints and on [unconstrained](#) models (see next slide);
 - ▶ this problem is worsened by the weak power of unit root and cointegration tests (against highly persistent alternative) and, in some models, by the non observability of the state variables;
- b) the "**bias problem**": the consequences of large finite sample biases occurring in [unconstrained estimators when the DGP is nearly non-stationary](#) and [near-cointegrated](#);
the key notion that is used is the notion of averaging estimators.
- c) the [application to a dynamic macro-finance Gaussian ATSM](#) highlights the relevance of this issue to extract a reliable term premia measure.

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K -step ahead short rate forecasts from the CVar(3) model; K from 1 to 40 quarters.



K -step ahead short rate forecasts from the VAR(3) model; K from 1 to 40 quarters.

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- ▶ To immediately highlight how this (downward) bias has dramatic consequences in terms of term premium estimates, we consider a simple illustration.
- ▶ Let us consider the following process for the short-term interest rate under the physical measure (monthly frequency):

$$i_{t+1} = \bar{i} + \phi(i_t - \bar{i}) + \sigma \varepsilon_{t+1}, \quad \varepsilon_t \sim \mathcal{N}(0, 1).$$

and the following under the risk-neutral measure:

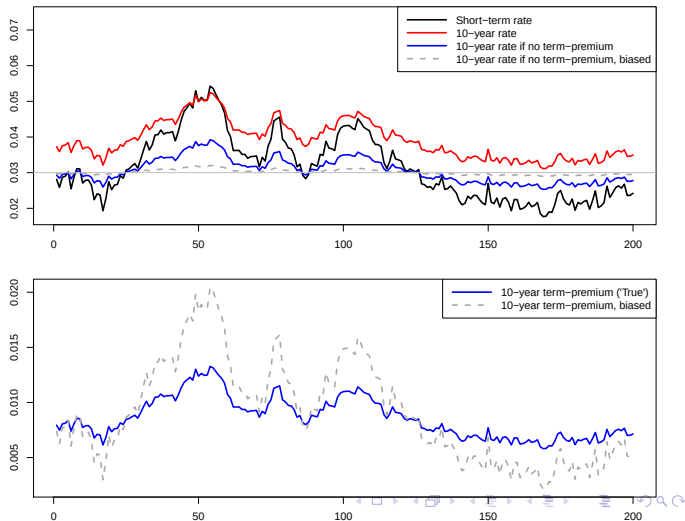
$$i_{t+1} = \bar{i}^* + \phi^*(i_t - \bar{i}^*) + \sigma \varepsilon_{t+1}, \quad \varepsilon_t \sim \mathcal{N}(0, 1).$$

with $\bar{i} = 3\%/12$, $\bar{i}^* = 6\%/12$, $\phi = 0.97$, $\phi^* = 0.99$, $\sigma = 0.2\%$.

- ▶ Assume the estimate of ϕ is downward biased ($\hat{\phi} = 0.9$). Influence on the 10-year term premium? [Figure 2, next slide]

I.6 – THE PERSISTENCE PROBLEM

Figure 2: Biased persistency: Implications on term premiums estimates



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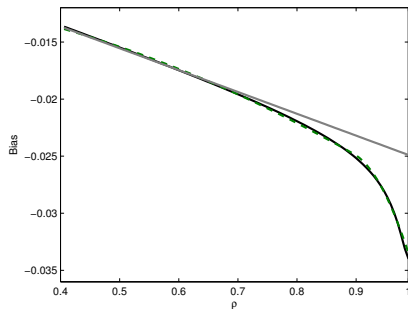
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I.6 – THE PERSISTENCE PROBLEM

Figure 4: Kendall's approximation $-(1 + 3\rho)/T$ and exact bias.



Bias of the OLS estimator $\hat{\rho}_T$ with sample size $T = 160$ (5×10^5 simulations). Exact (black solid line), Quadratic Spline approximation (green dashed line), Kendall's approximation (grey solid line).

I.6 – THE PERSISTENCE PROBLEM

Alternative approach: averaging estimators: proposed by B. Hansen (2010).

Idea:

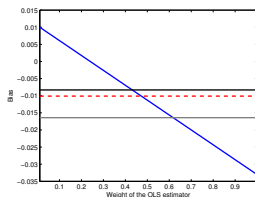
- ▶ use a weighted average of estimators taking and not taking into account non-stationarity aspects;
- ▶ using "local to unit root" asymptotics it can be shown, in a univariate setting, that these estimators have nice short-term prediction properties, and, in particular that the best weighting is strictly between 0 and 1.

Unfortunately:

- ▶ the optimal weight depends on the rate of convergence of the DGP to non-stationarity;
- ▶ generalization to a multivariate setting is difficult;

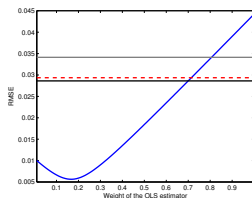
Here we consider the family estimators: $\hat{\rho}_T^A(\lambda) = (1 - \lambda) + \lambda\hat{\rho}_T$, $0 \leq \lambda \leq 1$.

Figure 5: Bias of estimators.



Bias of $\hat{\rho}_T^A$ (blue solid curve), $\hat{\rho}_T^I$ (black solid line), $\hat{\rho}_T^B$ (red dashed line)
and $\hat{\rho}_T^M$ (grey solid line), when $\rho = 0.99$ and $T = 160$.

Figure 6: RMSE of estimators.



RMSE of $\hat{\rho}_T^A$ (blue solid curve), $\hat{\rho}_T^I$ (black solid line), $\hat{\rho}_T^B$ (red dashed line)
and $\hat{\rho}_T^M$ (grey solid line), when $\rho = 0.99$ and $T = 160$.

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- ▶ Are the bias correction methods appropriate for predictions ?
Are the averaging estimators potentially more efficient ?
- ▶ By means of the following bivariate example, we evaluate and compare predictions in terms of **RMSFE ratio** (i.e. the RMSFE divided by the RMSFE obtained with the true ρ).
- **DGP :**

$$\begin{aligned}y_{1t} &= (1 - \rho) + \rho y_{1,t-1} + \varepsilon_{1t} \\y_{2t} &= 2y_{1t} + \varepsilon_{2t}\end{aligned}$$

with $\rho \in \{0.97, 0.98, 0.99\}$, where ε_{1t} , ε_{2t} independent standard Gaussian white noises.

- **Unconstrained VAR(1) model:**

$$y_t = \nu + Ay_{t-1} + \eta_t ,$$

and unconstrained OLS estimators of ν and A are $\hat{\nu}_T^{(u)}$ and $\hat{A}_T^{(u)}$.

I.6 – THE PERSISTENCE PROBLEM

- Error correction model (ECM) imposing one cointegrating relationship:

$$\Delta y_t = \mu + \alpha(y_{1,t-1} - \beta y_{2,t-1}) + \xi_t,$$

and associated constrained estimators of ν and A :

$$\hat{\nu}_T^{(c)} = \hat{\mu}_T \text{ and } \hat{A}_T^{(c)} = I + \hat{\alpha}_T(1, -\hat{\beta}_T),$$

where $\hat{\beta}_T$ obtained by regressing y_{1t} on y_{2t} , while $\hat{\mu}_T$ and $\hat{\alpha}_T$ are obtained by regressing Δy_t on $(1, y_{1,t-1} - \hat{\beta}_T y_{2,t-1})$.

- The **class of averaging estimators** is:

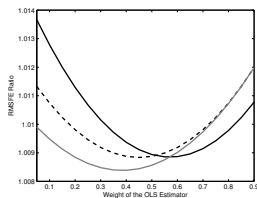
$$\hat{\nu}_T(\lambda) = (1 - \lambda)\hat{\nu}_T^{(c)} + \lambda\hat{\nu}_T^{(u)},$$

$$\hat{A}_T(\lambda) = (1 - \lambda)\hat{A}_T^{(c)} + \lambda\hat{A}_T^{(u)}, \quad 0 \leq \lambda \leq 1,$$

- and the **predictions of y_{T+h}** at T are :

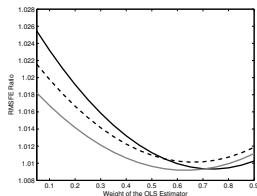
$$\hat{y}_{T,h}(\lambda) = [I - \hat{A}_T(\lambda)]^{-1}[I - \hat{A}_T^h(\lambda)]\hat{\nu}_T(\lambda) + \hat{A}_T^h(\lambda)y_T.$$

Figure 7: RMSFE ratios for y_{1t} when $h = 1$.



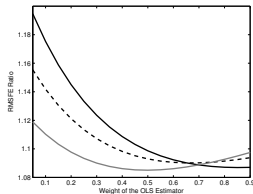
RMSFE ratio for y_{1t} when $T = 160$. $\rho = 0.97$ (black solid line),
 $\rho = 0.98$ (black dashed line), $\rho = 0.99$ (grey solid line).

Figure 8: RMSFE ratios for y_{2t} when $h = 1$.



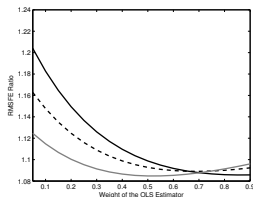
RMSFE ratio for y_{2t} when $T = 160$. $\rho = 0.97$ (black solid line),
 $\rho = 0.98$ (black dashed line), $\rho = 0.99$ (grey solid line).

Figure 9: RMSFE ratios for y_{1t} when $h = 20$.



RMSFE ratio for y_{1t} when $T = 160$. $\rho = 0.97$ (black solid line),
 $\rho = 0.98$ (black dashed line), $\rho = 0.99$ (grey solid line).

Figure 10: RMSFE ratios for y_{2t} when $h = 20$.



RMSFE ratio for y_{2t} when $T = 160$. $\rho = 0.97$ (black solid line),
 $\rho = 0.98$ (black dashed line), $\rho = 0.99$ (grey solid line).

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The general procedure proposed by JMP (2013) is the following:

- ▶ $\underline{y}_T = (y_1, \dots, y_T)$ = observations; $g(\underline{y}_t)$, $t \in \{1, \dots, T\}$ is a variable of interest that we want to predict at horizon h .
- define a sequence of increasing windows $\{1, \dots, t\}$, with $t \in \{t_0, \dots, T - h\}$;
- for each t compute $\hat{\theta}_t^{(u)}$ and $\hat{\theta}_t^{(c)}$ estimators of θ ;
- for each t compute $\hat{\theta}_t(\lambda) = (1 - \lambda)\hat{\theta}_t^{(c)} + \lambda\hat{\theta}_t^{(u)}$, the predictions $\hat{g}_{t,h}(\lambda)$ of $g(\underline{y}_{t+h})$ and the prediction error $g(\underline{y}_{t+h}) - \hat{g}_{t,h}(\lambda)$;

- compute $Q_T(\lambda, h) = \sum_{t=t_0}^{T-h} [g(\underline{y}_{t+h}) - \hat{g}_{t,h}(\lambda)]^2$;

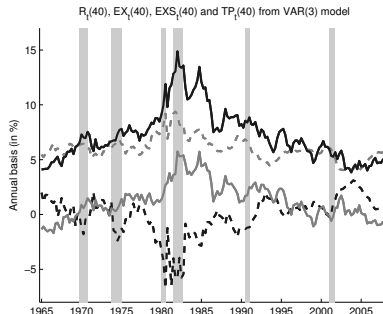
- calculate $\lambda^*(h) = \operatorname{argmin}_{\lambda \in [0,1]} Q_T(\lambda, h)$;
- compute $\hat{\theta}_T(\lambda^*(h))$.

I.6 – THE PERSISTENCE PROBLEM

The proposed Gaussian ATSM:

- ▶ $w_t = (\text{short rate, long rate, economic activity}) = (r_t, R_t, G_t)'$;
 $w_t \sim \text{VAR}(3)$;
- ▶ $\theta_{\mathbb{P}}$ are estimated by averaging estimator instead of OLS;
- ▶ Given the aim is to provide a reliable measure term premia on long term bonds, the paper focus on the minimization of the prediction error of the associated expectation part.
- ▶ $M_{t,t+1} = \exp \left[-r_{t+1} + \Gamma_t' \eta_{t+1} - \frac{1}{2} \Gamma_t' \Gamma_t \right]$ is the one-period SDF;
- ▶ $\Gamma_t = \gamma_0 + \gamma W_t$, $W_t = (w_t, w_{t-1}, w_{t-2})$;
- ▶ (γ_0, γ) estimated by CNLLS;
- ▶ $R_t(h) = -\frac{1}{h} \log B_t(h) = -\frac{c_h'}{h} W_t - \frac{d_h}{h}$, $h \geq 1$

Figure 11: Decomposition of 10-year yield: Gaussian ATSM with OLS-based VAR(3) model.



The black solid line denotes $R_t(40)$, $EX_t(40)$ (grey dashed line), $TP_t(40)$ (grey solid line) and $EXS_t(40)$ (black dashed line) measures from VAR(1) model. Shaded areas indicate recession periods (NBER).

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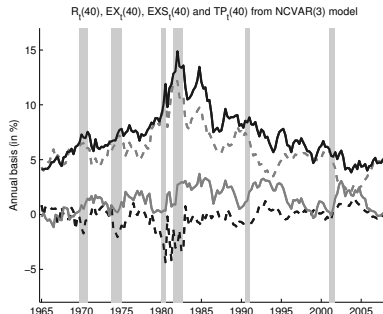
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Figure 12: Decomposition of 10-year yield: Gaussian ATSM with Averaging-Estimator-based VAR(3) model.



The black solid line denotes $R_t(40)$, $EX_t(40)$ (grey dashed line), $TP_t(40)$ (grey solid line) and $EXS_t(40)$ (black dashed line) measures from VAR(1) model. Shaded areas indicate recession periods (NBER).

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I.6 – THE PERSISTENCE PROBLEM

- ▶ Kim and Orphanides (2005) and Kim and Orphanides (2012) have proposed an alternative simple approach to deal with this problem.
- ▶ They add measurement equations to impose that the model-implied forecasts are –up to some measurement errors– close to survey-based ones.
- ▶ Kim and Orphanides use the Blue Chip Financial Forecasts. Alternative (publicly available) surveys: Philly Fed Survey of Professional Forecasters and ECB SPF].
- ▶ Importantly, this approach is tractable because model-implied forecasts of yields are affine in the state vector. [This is the case for any affine model.]
- ▶ Their model is able to fit both survey-based forecasts and market yields [Figure 13, next slide].

Figure 13: Fit of surveys

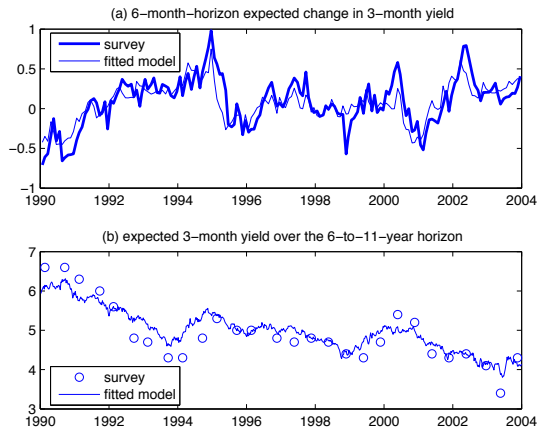


Figure 4: (a) Survey forecast of the 6-month-horizon expected change in the 3-month yield and the forecast implied by the model. (b) Survey forecast of the expected 3-month yield in the next 6 to 11 years and the model forecast.

Source: Kim and Orphanides (2005).

Appendix I.A (Kitagawa-Hamilton filter)

Consider a q -dimensional vector of variables F_t and an exogenous homogenous Markov chain z_t . We make use of the following notations:

- ▶ η_t is a J -dimensional vector whose j^{th} component is the p.d.f. of F_t conditional on $(z_t = e_j, \underline{F_{t-1}})$, i.e. $f(F_t | z_t = e_j, \underline{F_{t-1}})$
- ▶ ξ_t is a J -dimensional vector whose j^{th} component is $\mathbb{P}(z_t = e_j | \underline{F_t})$.

The sequence ξ_t can then be computed recursively as follows:

$$\xi_t = \frac{(\Pi' \xi_{t-1}) \odot \eta_t}{\mathbf{1}' (\Pi' \xi_{t-1} \odot \eta_t)}, \quad (I.19)$$

where $\odot$ denotes the element-by-element (Hadamard) product and where $\mathbf{1}$ denotes a J -dimensional vector of ones.

Moreover, the previous formulas also show how to compute the likelihood of the model since:

$$f(F_t | \underline{F_{t-1}}) = \mathbf{1}' (\Pi' \xi_{t-1} \odot \eta_t). \quad (I.20)$$

- The matrix of transition probabilities of $\mathcal{Z}_t$ is of the form $\mathbf{1}_{n \times 1} \otimes \tilde{\Pi}$, with

$$\tilde{\Pi} = \begin{bmatrix} \pi_{1,\bullet} & 0_{1 \times n} \dots & & & 0_{1 \times n} \\ 0_{1 \times n} & \pi_{2,\bullet} & 0_{1 \times n} & \dots & 0_{1 \times n} \\ & & \ddots & & \\ & & 0_{1 \times n} & \pi_{n-1,\bullet} & 0_{1 \times n} \\ 0_{1 \times n} & \dots & & 0_{1 \times n} & \pi_{n,\bullet} \end{bmatrix},$$

where $\pi_{i,\bullet}$ denotes the i^{th} row of Π (Π is defined on Slide 37).

- The last term appearing in Eq. (I.18) can be computed as follows:

$$\begin{aligned} & f \left(\begin{bmatrix} x(y_t, z(\mathcal{Z}_t)) \\ y_{2,t} \end{bmatrix} \middle| \mathcal{Z}_t, \underline{y_{t-1}} \right) \\ &= f(y_{2,t} | x_t = x(y_t, z(\mathcal{Z}_t)), \mathcal{Z}_t, \underline{y_{t-1}}) \times \\ & \quad f(x(y_t, z(\mathcal{Z}_t)) | \mathcal{Z}_t, \underline{y_{t-1}}) \\ &= \mathbf{n}(y_{2,t}; A_2 z_t + B_2 x_t, \Omega_2) \times \\ & \quad f_\varepsilon(\varepsilon_t | z_t = z(\mathcal{Z}_t), x_{t-1} = x(y_{t-1}, z_{-1}(y_{t-1}, \mathcal{Z}_t))), \end{aligned}$$

where $\mathbf{n}$ is the p.d.f. of a multivariate normal distri, (Prop. I.3), where $\varepsilon_t = x_t - m[z(\mathcal{Z}_t), x(y_{t-1}, z_{-1}(y_{t-1}, \mathcal{Z}_t))]$ (m defined in Eq. I.13).

Chapter II

CREDIT AND LIQUIDITY RISKS

– Agenda –

- II.1 Notations for an introductory example. (Slide 70)
- II.2 General notations. (Slide 80)
- II.3 Pricing under standard assumptions. (Slide 81)
- II.4 Pricing illiquid bonds. (Slide 95)
- II.5 Relaxing the classical assumptions. (Slide 109)
- II.6 General affine credit-risk framework. (Slide 111)
- II.7 CDS Pricing in the general framework. (Slide 119)
- II.8 Top-Down Approach. (Slide 125)
 - ▶ Appendix II.A: Structural interpretation of the illiquidity loss.
 - ▶ Appendix II.B: Proof of Prop. II.1.
 - ▶ Appendix II.C: General pricing under the RFV convention.

Dynamic credit risk models: Why?

- ▶ Compute Values-at-Risk of portfolios of credit-sensitive instruments.
- ▶ Extract term structures of default probabilities from credit spreads (CDS).
- ▶ Compute expected excess returns of defaultable bonds.
- ▶ Price exotic credit derivatives.
- ▶ Academic perspective: Understanding the pricing of credit risk.

II.1 – Notations for an introductory example

- How do we properly take into account credit risk (default event) in bond pricing ?
- Let us consider the *old problem* to price at date t a ZCB maturing at date $T > t$ (with unitary face value).
- Now this asset is issued by a firm characterized by a positive probability of default and its price is denoted $P_z^{(d)}(t, T)$.
- It is clear that $P_z^{(d)}(t, T)$ is going to also depend on:
 - i)* probability that the firm make default before T and on
 - ii)* the recovery rate in case of default.
- τ denotes the **random time of default**; if $\tau > T \Rightarrow$ no default.

II.1 – Notations for an introductory example

□ τ is a non-negative random variable characterized by an *Exponential Law* with parameter λ :

i) $f_{\lambda}(\tau) = \lambda \exp(-\lambda \tau) \mathbb{I}_{\{\tau \geq 0\}}$, $\mathbb{E}(\tau) = \frac{1}{\lambda}$ and $\mathbb{V}(\tau) = \frac{1}{\lambda^2}$

ii) $\mathbb{P}(\tau \leq t) = 1 - \exp(-\lambda t)$, thus $\mathbb{P}(\tau > t) = \exp(-\lambda t)$

iii) lack of memory:

$$\begin{aligned}\mathbb{P}(\tau > t + s | \tau > t) &= \frac{\mathbb{P}[\{\tau > t + s\} \cap \{\tau > t\}]}{\mathbb{P}(\tau > t)} \\ &= \frac{\mathbb{P}(\tau > t + s | \tau > t + s) \mathbb{P}(\tau > t + s)}{\mathbb{P}(\tau > t)} \\ &= \frac{\mathbb{P}(\tau > t + s)}{\mathbb{P}(\tau > t)} = \exp(-\lambda s).\end{aligned}$$

iv) $\lim_{s \rightarrow 0} \frac{\mathbb{P}(\tau > t + s | \tau > t)}{s} = \lambda \quad \forall \quad t > 0$, and λ is seen as the **default intensity parameter**.

II.1 – Notations for an introductory example

- Given the default time τ , we define (again) the default process by:

$$d_{\tau, T} = \mathbb{1}_{\{\tau \leq T\}} = \begin{cases} 1 & \text{if default at } \tau \leq T \\ 0 & \text{if not.} \end{cases}$$

It is a point process with one jump of size one at default.

- RR_T denotes the recovery rate and it is constant.
- The default event can happen at dates $t_1 < t_2 < \dots < t_N = T$, but the recovery is anyway at T .
- We assume a risk-free discount factor with a constant risk-free short rate ($r_t = r$): $Z(0, t) = \exp(-r \times t)$.

II.1 – Notations for an introductory example

□ Let us assume to be at date $t \in (0, T)$ and the firm is alive.

- We assume that the probability that the firm make default by time T , given that we know that at t is alive is given by:

$$\mathbb{P}(\tau \leq T \mid \tau > t) = \mathbb{P}(\tau \leq T - t) = 1 - e^{-\lambda(T-t)} = DP_{\tau,t,T}$$

- The **survival probability** $\mathcal{S}_{\tau,t,T}$ is given by:

$$\mathcal{S}_{\tau,t,T} = \mathbb{P}(\tau > T \mid \tau > t) = e^{-\lambda(T-t)} = 1 - \mathbb{P}(\tau \leq T - t)$$

- The unitary payoff of the defaultable ZCB is:

$$\begin{aligned} y(T) &= RR_T \mathbb{1}_{\{\tau \leq T\}} + \mathbb{1}_{\{\tau > T\}} &= & RR_T d_{\tau,T} + (1 - d_{\tau,T}) \\ & &= & 1 - d_{\tau,T} (1 - RR_T). \end{aligned}$$

II.1 – Notations for an introductory example

- Its price at t , given $\tau > t$, is:

$$\begin{aligned} & P_z^{(d)}(t, T) \\ &= e^{(-r(T-t))} \times \mathbb{E}_t[1 - d_{\tau, T} (1 - RR_T) | \tau > t] \\ &= e^{(-r(T-t))} - e^{(-r(T-t))} (1 - RR_T) \mathbb{E}_t \left[\mathbb{1}_{\{\tau \leq T\}} | \tau > t \right] \\ &= e^{(-r(T-t))} - e^{(-r(T-t))} (1 - RR_T) \mathbb{P}(\tau \leq T | \tau > t) \\ &= e^{(-r(T-t))} - e^{(-r(T-t))} (1 - RR_T) DP_{\tau, t, T}. \end{aligned}$$

where $P_z(t, T) = \exp(-r(T-t))$ is the non-defaultable ZCB price and the second term is the **risk-free discounted value of the expected loss**.

- Indeed, we have an Exposure At Default of $EAD_T = 1$, a Loss Given Default given by $LGD_T = (1 - RR_T)$, and $\mathbb{E}_t(d_{\tau, T}) = DP_{\tau, t, T}$.
- Thus, the loss at date T is $L_T = (1 - RR_T) d_{\tau, T}$ and therefore $\mathbb{E}_t(L_T) = (1 - RR_T) DP_{\tau, t, T}$.

II.1 – Notations for an introductory example

- We have that the difference between the risk-free (non defaultable) and the risky (defaultable) bond price is (in that simple setting) given by

$$\exp(-r(T-t)) \times (1 - RR_T) \times DP_{\tau,t,T},$$

that is the present value of the expected loss (associated to one unit of money).

- Observe that, if $RR_T = 0$, then $y(T) = 1 - d_{\tau,T}$ and

$$\begin{aligned} P_z^{(d)}(t, T) &= \exp(-r(T-t))(1 - DP_{\tau,t,T}) \\ &= \exp[-(r + \lambda) \times (T - t)], \end{aligned}$$

that is, the unitary payoff is discounted by $r_d = r + \lambda$, the risk-free rate increased by the **default intensity parameter** λ ; r_d is called **default risk adjusted** short rate.

- If $DP_{\tau,t,T} = 0$ (or $RR_T = 1$) then $P_z^{(d)}(t, T) = P_z(t, T)$.
- The ZCB price $P_z^{(d)}(t, T)$ **increases** as $DP_{\tau,t,T}$ **decreases**
- The ZCB price $P_z^{(d)}(t, T)$ **increases** as RR_T **increases** toward one.

II.1 – Notations for an introductory example

- Remember the assumptions :
 - recovery and short rates are assumed constant (no uncertainty);
 - default time τ , recovery rate (and discount factors) are independent;
- This simple setting has highlighted the fact that, even if we ignore the uncertainty about:
 - future spot rates and recovery rates,
 - as well as the dependence between τ , RR_T and r_t ,

the simple presence of the default event, possibly affecting the payoff we have to price, ask for a larger discount rate $r + \lambda$.

- Before moving to the following section, let us consider a small generalization of the previous setting:
 - we still have perfect foresight about the future sequence of (non constant!) risk-free short rates;
 - in case of default at any date $\tau \leq T$, the recovery takes place at T ;
 - default time τ and RR_T (and short rates) are independent.

II.1 – Notations for an introductory example

- The payoff at T of the defaultable ZCB is given by:

$$\begin{aligned}y(T) &= \left[RR_T \mathbb{1}_{\{\tau \leq T\}} + \mathbb{1}_{\{\tau > T\}} \right] \times 100 \\&= \left[RR_T d_{\tau, T} + (1 - d_{\tau, T}) \right] \times 100 \\&= \left[1 - d_{\tau, T} (1 - RR_T) \right] \times 100.\end{aligned}$$

- The price at date t is

$$\begin{aligned}&P_z^{(d)}(t, T) \\&= e^{(-r(t, T)(T-t))} \times \mathbb{E}_t \left[\left(1 - d_{\tau, T} (1 - RR_T) \right) 100 \mid \tau > t \right] \\&= P_z(t, T) - P_z(t, T) \mathbb{E}_t \left[\mathbb{I}_{\{\tau \leq T\}} (1 - RR_T) \mid \tau > t \right] \\&= P_z(t, T) - P_z(t, T) \mathbb{E}_t \left[(1 - RR_T) \right] \mathbb{P}(\tau \leq T \mid \tau > t) \\&= P_z(t, T) - P_z(t, T) \mathbb{E}_t \left[(1 - RR_T) \right] DP_{\tau, t, T} \\&= P_z(t, T) \times \left\{ 1 - DP_{\tau, t, T} \times [1 - \mathbb{E}_t(RR_T)] \right\}\end{aligned}$$

II.1 – Notations for an introductory example

- Using the notation $r_t = r(t, t+1)$ and assuming that future short rates are unknown now, we can write (further small generalization):

$$P_z^{(d)}(t, T) = \mathbb{E}_t \left[\exp \left(- \sum_{i=t}^{T-1} r_i \right) \times (1 - d_{\tau, T} (1 - RR_T)) \times 100 \mid \tau > t \right]$$

and if $RR_T = 0$ we have:

$$\begin{aligned} P_z^{(d)}(t, T) &= \mathbb{E}_t \left[\exp \left(- \sum_{i=t}^{T-1} r_i \right) \times (1 - d_{\tau, T}) \times 100 \mid \tau > t \right] \\ &= \mathbb{E}_t \left[\exp \left(- \sum_{i=t}^{T-1} r_i \right) \times \mathbb{I}_{\{\tau > T\}} \mid \tau > t \right] \times 100 \\ &= \mathbb{E}_t \left[\exp \left(- \sum_{i=t}^{T-1} r_i \right) \times 1 \mid \tau > T, \tau > t \right] \mathbb{P}[\tau > T \mid \tau > t] \times 100 \\ &\quad + \mathbb{E}_t \left[\exp \left(- \sum_{i=t}^{T-1} r_i \right) \times 0 \mid \tau \leq T, \tau > t \right] \mathbb{P}[\tau \leq T \mid \tau > t] \times 100 \end{aligned}$$

II.1 – Notations for an introductory example

□ Thus, we have:

$$\begin{aligned}P_z^{(d)}(t, T) &= \mathbb{E}_t \left[\exp \left(- \sum_{i=t}^{T-1} r_i \right) \mid \tau > T, \tau > t \right] \exp [-\lambda (T - t)] \times 100 \\&= \mathbb{E}_t \left\{ \exp \left[- \sum_{i=t}^{T-1} (r_i + \lambda) \right] \right\} \times 100 \\&= \mathbb{E}_t \left\{ \exp \left[- \sum_{i=t}^{T-1} r_{d,i} \right] \right\} \times 100 .\end{aligned}$$

- We observe that we are no more able to **isolate** the probability of default within the pricing formula.
- These pricing formulas are still implicitly assuming a risk premium equal to zero given that we are discounting a risky payoff with the risk-free rate.
- The risk-free rate is adjusted only by the default event (λ) but not about the future rates uncertainty, the links with future recovery rate, default event and the economy.

II.2 – Notations

- The information of the investor at date t is $\underline{w}_t = (w'_t, w'_{t-1}, \dots, w'_1)'$, where $w_t = (y'_t, d'_t)'$:
- y_t is a n_y -dimensional vector of common factors,
 - $d_t = (d_{1,t}, \dots, d_{E,t})'$ a E -dimensional vector of binary variables representing the possible default of entities $e \in \{1, \dots, E\}$:
 $d_{e,t} = 1 \Leftrightarrow$ entity e is in default status at time t .
 $d_{e,t} = 0 \Leftrightarrow$ entity e is not in default status at time t .
 - $\tau^{(e)}$ is the default date and $d_{e,t} = \mathbb{1}_{\{\tau^{(e)} \leq t\}}$.
- w_t is a K -dimensional vector $\Rightarrow K = n_y + E$. We assume that the process $\{w_t\}$ is Markov $\Rightarrow$ Its historical ($\mathbb{P}$) dynamics is defined by the conditional densities:

$$f(y_t | w_{t-1}) \text{ and } p(d_t | y_t, w_{t-1}).$$

- We also introduce a one-period positive SDF between $t-1$ and t , denoted $M_{t-1,t}(w_t, w_{t-1})$.
- We define the **survival probability** as:

$$p_e(0 | y_t, w_{t-1}, d_{e,t-1} = 0) = \exp[- \underbrace{\lambda_e(y_t, w_{t-1})}_{\text{default intensity}}]. \quad (\text{II.1})$$

where $\lambda_e(y_t, w_{t-1})$ is a non-negative function.

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Assumption II.1 (absorbing defaults)

The default event is an absorbing state: $p_e(0 | y_t, w_{t-1}, d_{e,t-1} = 1) = 0$.

Assumption II.2 (y_t is exogenous or autonomous / non-systemic entity / no-jump condition)

$\{d_t\}$ does not Granger-cause $\{y_t\}$, in the sense that:

$$f(y_t | w_{t-1}) = f(y_t | y_{t-1}).$$

Assumption II.3 (unpriced credit events)

The default events (or credit events) are not priced, in the sense that:

$$M_{t-1,t}(w_t, w_{t-1}) = M_{t-1,t}(y_t, y_{t-1}).$$

Remark II.1 (Short-term risk-free rate)

The previous two assumptions notably imply that $r_t (= -\log \mathbb{E}_t(M_{t,t+1}))$ depends on $\underline{y}_t$ (but not on d_t).

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

- ▶ We start with a presentation of the **classical credit-risk framework**, and let us first focus on a single entity ($d_t \equiv d_{1,t}$), alive on date t ($d_t = 0$).

We have $d_t = \mathbb{1}_{\{\tau \leq t\}}$.

- ▶ Consider the date- t price of a bond issued by the defaultable entity; residual maturity h and zero recovery rate (0 payoff in case of default and paid at $t + h$):

$$B(t, h) = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{t+h-1})(1 - d_{t+h})].$$

- ▶ Conditioning w.r.t. $\underline{y_{t+h}}$, this price is given by:

$$\begin{aligned} & \mathbb{E}_t^{\mathbb{Q}}\{\mathbb{E}_t^{\mathbb{Q}}\{\exp(-r_t - \dots - r_{t+h-1})(1 - d_{t+h})|\underline{y_{t+h}}\}\} \\ &= \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{t+h-1})\mathbb{E}_t^{\mathbb{Q}}\{(1 - d_{t+h})|\underline{y_{t+h}}\}] \\ &= \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \dots - r_{t+h-1})\mathbb{Q}\{d_{t+h} = 0|\underline{y_{t+h}}\}] \\ &= \mathbb{E}_t^{\mathbb{Q}}\left[\exp(-r_t - \dots - r_{t+h-1})\prod_{j=1}^h \mathbb{Q}\{d_{t+j} = 0|d_{t+j-1} = 0, \underline{y_{t+h}}\}\right]. \end{aligned}$$

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

- By Bayes, we have:

$$f^{\mathbb{Q}}(d_t | d_{t-1}, y_t, y_{t-1}) = \frac{f^{\mathbb{Q}}(d_t, y_t | d_{t-1}, y_{t-1})}{f^{\mathbb{Q}}(y_t | d_{t-1}, y_{t-1})}.$$

- Using Eq. (II.3) (numerator) and Eq. (II.4) (denominator), we get:

$$f^{\mathbb{Q}}(d_t | d_{t-1}, y_t, y_{t-1}) = f^{\mathbb{P}}(d_t | d_{t-1}, y_t, y_{t-1}) \Rightarrow \lambda_t^{\mathbb{Q}} = \lambda_t^{\mathbb{P}}.$$

- Using this in Eq. (II.2), we obtain that, under Assumptions II.2 and II.3, and for a zero recovery rate:

$$B(t, h) = \mathbb{E}_t^{\mathbb{Q}} [\exp(-r_t - \dots - r_{t+h-1} - \lambda_{t+1} - \dots - \lambda_{t+h})]. \quad (\text{II.5})$$

Remark II.2

Although $\mathbb{P}$ and $\mathbb{Q}$ intensities are the same functions of y_t and y_{t-1} , their historical and risk-neutral dynamics are in general different since y_t 's $\mathbb{P}$ and $\mathbb{Q}$ dynamics are different.

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

- Eq. (II.5) rewrites:

$$B(t, h) = \mathbb{E}_t^{\mathbb{Q}} [\exp(-\tilde{r}_t - \cdots - \tilde{r}_{t+h-1})], \quad (\text{II.6})$$

where $\tilde{r}_t = r_t + \lambda_{t+1}$.

- Eq. (II.6) is reminiscent of ZCB pricing equation we have seen in the ATSM literature (previous lectures), where the risk-free short-term rate r_t is replaced by the **credit-adjusted short-term rate** $\tilde{r}_t$.
- If both r_t and λ_t are affine in y_t , where the latter is an affine process, then bond prices are easily computed by means of recursive formulas (as was the case in the context risk-free interest rates).
- In this case, $B(t, h)$ is exponential affine in y_t , that is, it is of the form:

$$B(t, h) = \exp(A_h^{d'} y_t + B_h^d) \quad (\text{say}). \quad (\text{II.7})$$

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

- ▶ The previous framework can be adapted in order to accommodate non-zero recovery rates. In this framework, the default intensity will be replaced by a **pseudo**, or **recovery-adjusted default intensity** $\tilde{\lambda}_t$.

- ▶ **Recovery of Market Value [RMV]** modelling convention:

*In case of default, the recovery payoff is a fraction $\zeta \in [0, 1]$ (**Recovery Rate, RR**) of the “price that would have prevailed, absent the default”.*

- ▶ **Recovery of Face Value [RFV]** modelling convention:

In case of default at $t + i$, the recovery payment is given by a fraction $\zeta \in [0, 1]$ of the (unitary) face value.

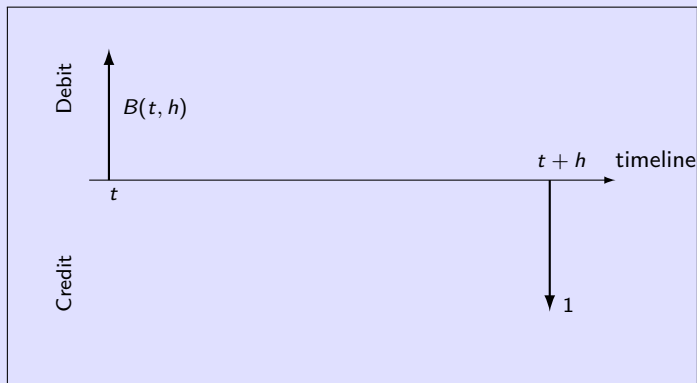
- ▶ **Recovery of Treasury [RT]** modelling convention:

In case of default at date $t + i$, the recovery payment is $\zeta B_{rf}(t + i, h - i)$, where $B_{rf}(t, h)$ is the date- t market price of a risk-free ZCB maturing at $t + h$.

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Payoffs of a zero-coupon issued by e (when no default)

Figure 14: At t , investor buys a bond issued by e , maturity date $t + h$



Estimation of
Affine
Asset-Pricing
Models

Pricing Credit and
Liquidity Risks

Notations for an
introductory example

General notations

Pricing Under Standard
Assumptions

Pricing Illiquid Bonds

Relaxing the Classical
Framework Assumptions

General Affine Credit-Risk
Framework

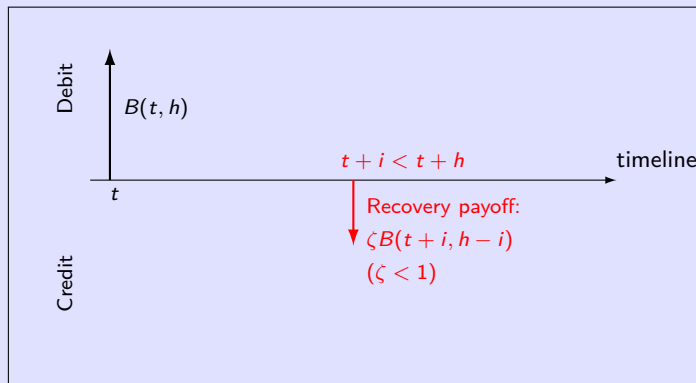
CDS Pricing in the General
Framework

Top-Down Approach

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Payoffs of a zero-coupon issued by e (default on $t + i$, RMV convention)

Figure 15: At t , investor buys a bond issued by e , maturity date $t + h$



II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Sketch of the proof of Eq. (II.9)

- If $d_t = 0$ (entity alive on date t), we then have:

$$\begin{aligned} B(t, h) &= \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}} [\underbrace{d_{t+1} \zeta B(t+1, h-1)}_{\text{payoff under default}} + \underbrace{(1-d_{t+1}) B(t+1, h-1)}_{\text{payoff without default}}] \\ &= \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}} [\{1 - d_{t+1}(1 - \zeta)\} B(t+1, h-1)]. \end{aligned}$$

- Defining the **recovery-adjusted default intensity** $\tilde{\lambda}_{t+1}$ through

$$\begin{aligned} \exp(-\tilde{\lambda}_{t+1}) &= \mathbb{E}[\{1 - d_{t+1}(1 - \zeta)\} | \underline{y}_{t+1}, d_t = 0] \\ &= 1 - (1 - \zeta)(1 - \exp(-\lambda_{t+1})), \end{aligned} \quad (\text{II.8})$$

one obtains:

$$B(t, h) = \mathbb{E}_t^{\mathbb{Q}} [\exp(-r_t - \tilde{\lambda}_{t+1}) B(t+1, h-1)],$$

and, by recursion:

$$B(t, h) = \mathbb{E}_t^{\mathbb{Q}} [\exp(-r_t - \tilde{\lambda}_{t+1} - \dots - r_{t+h-1} - \tilde{\lambda}_{t+h})]. \quad (\text{II.9})$$

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

- This is formalized in the following proposition.

Prop. II.1 (Bond pricing in the RMV classical framework)

Under Assumptions II.2 and II.3, and if, in case of default on date $t + i$, the recovery payment is $\zeta \tilde{B}(t + i, h - i)$, where:

- the **pseudo-price** $\tilde{B}(t, h)$ is given by:

$$\tilde{B}(t, h) = \mathbb{E}_t^{\mathbb{Q}}[\exp(-r_t - \tilde{\lambda}_{t+1} - \cdots - r_{t+h-1} - \tilde{\lambda}_{t+h}), h \geq 1,$$

with $\tilde{B}(t, 0) = 1$.

- the *pseudo-intensity* $\tilde{\lambda}_t$ is defined in Eq. (II.8),

- $\zeta \in [0, 1]$,

then

$$B(t, h) = (1 - d_t)\tilde{B}(t, h). \quad (\text{II.10})$$

Proof. See Appendix II.B. ■

- Again, if r_t and $\tilde{\lambda}_t$ are affine combinations of an affine process y_t , then (exponential affine) bond prices are easily computed by means of recursive formulas, see on next slide.

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Assumption II.4

The process $\{y_t\}$ is affine under the $\mathbb{Q}$ measure:

$$\varphi_{y,t-1}^{\mathbb{Q}}(u_y) = \mathbb{E}^{\mathbb{Q}} \left[\exp(u'_y y_t) \mid \underline{y}_{t-1} \right] = \exp \left[a_y^{\mathbb{Q}}(u_y)' y_{t-1} + b_y^{\mathbb{Q}}(u_y) \right], \quad (\text{II.11})$$

and r_t and $\tilde{\lambda}_{e,t}$ are affine in (y_t, y_{t-1}) , that is (say):

$$r_t = \omega_0 + \omega'_1 y_t, \quad \tilde{\lambda}_{e,t} = \kappa_0 + \kappa'_1 y_t.$$

Prop. II.2 (Standard affine bond pricing, under RMV convention)

Under the Assumptions of Proposition II.1 and under Assumption II.4, we have, for $h \geq 1$:

$$B(t, h) = (1 - d_t) \exp(-h[\omega_0 + \kappa_0] - \omega'_1 y_t + A'_h y_t + B_h), \quad (\text{II.12})$$

where A_h and B_h are given by the following recursive equations:

$$\begin{cases} A_h &= a(u_h + A_{h-1}), \\ B_h &= b(u_h + A_{h-1}) + B_{h-1}, \\ A_0 &= 0, \quad B_0 = 0, \end{cases} \quad (\text{II.13})$$

with $u_1 = -\kappa_1$ and $u_i = -(\kappa_1 + \omega_1)$ for $i > 1$.

Proof. The results directly follows from Proposition II.1.

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

- ▶ What about when different entities are considered ($E = 2$)?
- ▶ Same pricing formulas are obtained under the assumption of no-contagion (Assumption II.5).

Assumption II.5 (no contagion)

There is no instantaneous or lagged contagion between entities, i.e.:

$$p(d_t | y_t, w_{t-1}) = p_1(d_{1,t} | y_t, y_{t-1}, d_{1,t-1}) \times p_2(d_{2,t} | y_t, y_{t-1}, d_{2,t-1}).$$

- ▶ If Assumption II.5 is not satisfied, e.g. if:

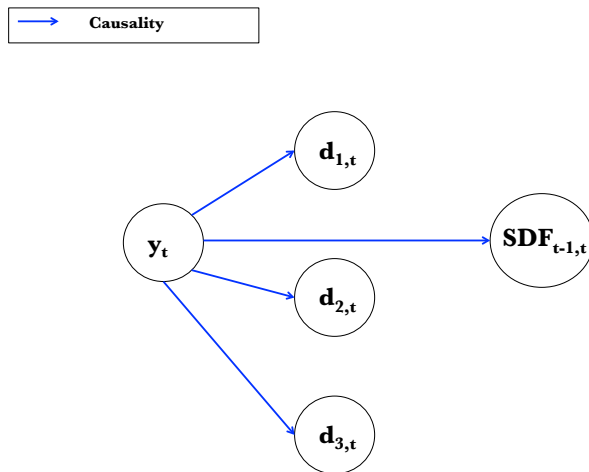
$$\begin{aligned} p(d_t | y_t, w_{t-1}) &= p_1(d_{1,t} | y_t, y_{t-1}, d_{1,t-1}) \times \\ &\quad p_2(d_{2,t} | y_t, y_{t-1}, d_{1,t}, d_{1,t-1}, d_{2,t-1}). \end{aligned}$$

(contagion effect from $e = 1$ towards $e = 2$), then:

- Proposition II.1 is still valid for both entities;
- In an affine framework, the computation of $\tilde{B}_1(t, h)$ (and $B_1(t, h)$) is straightforward;
- but formulas for $\tilde{B}_2(t, h)$ and $B_2(t, h)$ are not explicit anymore even if $\tilde{\lambda}_{2,t}$ is affine in $(y_t, y_{t-1}, d_{1,t}, d_{1,t-1})$.

II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Figure 16: Causality scheme in the classical credit-risk framework



II.3 – PRICING UNDER STANDARD ASSUMPTIONS

Prop. II.3

Under Assumptions II.1, II.2, II.3 and II.5, the risk-neutral ($\mathbb{Q}$, say) dynamics is such that:

$$f^{\mathbb{Q}}(y_t | w_{t-1}) = f^{\mathbb{Q}}(y_t | y_{t-1}) \propto M_{t-1,t}(y_t, y_{t-1}) f(y_t | y_{t-1})$$

(exogeneity of y_t preserved under $\mathbb{Q}$) and

$$p_t^{\mathbb{Q}}(d_t | y_t, w_{t-1}) = p_1(d_{1,t} | y_t, y_{t-1}, d_{1,t-1}) \times p_2(d_{2,t} | y_t, y_{t-1}, d_{2,t-1}),$$

(absence of contagion preserved under $\mathbb{Q}$).

*Denoting by $\lambda_e^{\mathbb{Q}}(y_t, y_{t-1}) \equiv -\log [p_e^{\mathbb{Q}}(0 | y_t, y_{t-1}, 0)]$ the **risk-neutral default intensity** of entity e , it comes that:*

$$\lambda_e^{\mathbb{Q}}(y_t, y_{t-1}) = \lambda_e(y_t, y_{t-1})$$

(default intensities are the same under $\mathbb{P}$ and $\mathbb{Q}$).

Proof. The $\mathbb{Q}$ conditional density of w_t given w_{t-1} , namely $f^{\mathbb{Q}}(y_t | w_{t-1}) p^{\mathbb{Q}}(d_t | y_t, w_{t-1})$, is proportional to $M_{t-1,t}(w_t, w_{t-1}) f(y_t | w_{t-1}) p(d_t | y_t, w_{t-1})$, and thus proportional to $M_{t-1,t}(y_t, y_{t-1}) f(y_t | y_{t-1}) p_1(d_{1,t} | y_t, y_{t-1}, d_{1,t-1}) p_2(d_{2,t} | y_t, y_{t-1}, d_{2,t-1})$. The result follows immediately.

II.4 – PRICING ILLIQUID BONDS (OR ASSETS)

- ▶ A very similar approach can be used to price illiquid assets.
- ▶ The idea consists in replacing the Loss Given Default ($LGD = 1 - RR^{(e)}$) by a Loss Given Liquidity Shock ($1 - \theta_\ell^{(e)}$).
- ▶ Structural interpretation building on **Ericsson and Renault (2006)** (see Appendix II.A).
- ▶ Novel binary variable: $\ell_t = \text{“liquidity shocks”}$
 $\ell_t = 1$ when the bondholder is hit, $\ell_t = 0$ otherwise.
- ▶ Liquidity shock = cash constraint or need to rebalance portfolio.
- ▶ **Liquidity-shock intensity** $\lambda_\ell(y_t)$, defined through:

$$\underbrace{1 - \exp[-\lambda_\ell(y_t)]}_{\approx \lambda_\ell(y_t) \text{ if small}} = \mathbb{P}(\ell_t = 1 | \underline{w}_{t-1}, y_t).$$

- ▶ ℓ_t is assumed not to Granger-cause y_t (as in Assumption II.2) and not to be priced (as in Assumption II.3). Using the same reasoning as in Slide 84, this implies in particular that

$$\lambda_\ell^{\mathbb{Q}}(y_t) = \lambda_\ell(y_t),$$

where

$$1 - \exp[-\lambda_\ell^{\mathbb{Q}}(y_t)] = \mathbb{Q}(\ell_t = 1 | \underline{w}_{t-1}, y_t).$$

II.4 – PRICING ILLIQUID BONDS (OR ASSETS)

Econometrics of Commodity and Asset Pricing

Lecture 5

Monfort &
Pegoraro & Renne

Notations for an introductory example

Pricing Under Standard Assumptions

Pricing Illiquid Bonds

Relaxing the Classical Framework Assumptions

General Affine Credit-Risk Framework

CDS Pricing in the General Framework

Top-Down Approach

- **Pseudo-illiquidity intensity** (mimics Eq. 11.8):

$$\begin{aligned} \tilde{\lambda}_{e,\ell,t}(y_t, y_{t-1}) &= -\log \left\{ \exp \left[-\lambda_{e,\ell}(y_t) \right] + \right. & (11.14) \\ &\quad \left. \left(1 - \exp \left[-\lambda_{e,\ell}(y_t) \right] \right) \theta_{\ell}^{(e)} \right\} \end{aligned}$$

- Adjusted **pseudo-price** for any date t and maturity h (with $\tilde{B}_e(t, 0) = 1$) (mimics Eq. II.9, replace):

$$\tilde{B}_e(t, h) = \mathbb{E}_t^{\mathbb{Q}} \left\{ \exp \left[- \sum_{i=1}^h \left(r_{t+i-1} + \tilde{\lambda}_{e,c,t+i} + \tilde{\lambda}_{e,\ell,t+i} \right) \right] \right\}. \quad (\text{II.15})$$

where $\tilde{\lambda}_{e,c,t+i}$ and $\tilde{\lambda}_{e,\ell,t+i}$ are the pseudo-credit and pseudo-illiquidity intensities.

- ▶ Upon the arrival of the liquidity shock ($\ell_t = 1$), the investor has to exit by selling her bond holdings. She is then limited in the number of traders she can call (see Appendix II.A) and, as a result, get only a fraction of the price she would get in “normal-liquidity times” ($\ell_t = 0$).

II.4 – PRICING ILLIQUID BONDS (OR ASSETS)

- Formally, consider a zero-coupon bond whose residual maturity on date t is h :

- If entity e defaults on date t , or if a liquidity shock hits the investor on date t , then the investor recovers:

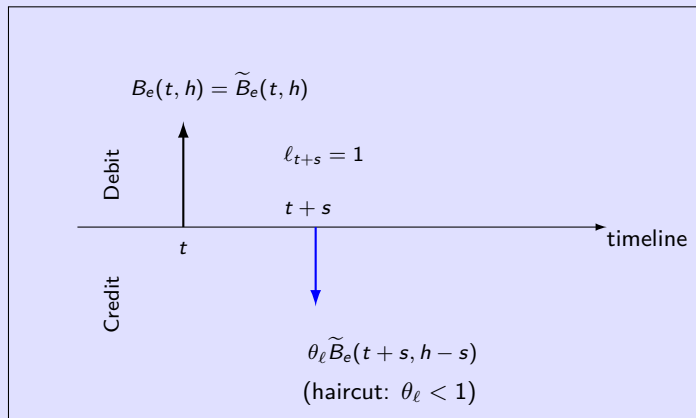
$$\left(1 - (1 - RR_t^{(e)})d_t^{(e)}\right) \left(1 - (1 - \theta_\ell^{(e)})\ell_t\right) \tilde{B}_e(t, h).$$

- Otherwise (from the investor point of view):

$$B_e(t, h) = (1 - d_t^{(e)})(1 - \ell_t)\tilde{B}_e(t, h).$$

- If the pseudo-illiquidity intensity $\tilde{\lambda}_{\ell,t}$ is an affine function of y_t , Prop. II.2 can easily be extended to the pricing of defaultable and illiquid bonds.

Payoffs of an illiquid bond (occurrence of liquidity shock)

Figure 17: At date $t + s$, the bondholder is hit by a liquidity shock

- ▶ *Monfort and Renne (MR2011) and Monfort and Renne (MR2014): credit/liquidity decompositions of euro-area sovereign spreads.*
- ▶ *Different estimation approaches: Kim (1994)'s filter (Slide 40) for MR2014, and Kitagawa-Hamilton filter (Slide 39) for MR2011.*
- ▶ $w_t = (y'_t, z'_t)'$ follows a Regime-Switching Gaussian VAR (see Example I.1).
- ▶ z_t (homogenous and exogenous Markov chain) = "crisis indicator".
- ▶ Specification of the **total intensity**:

$$\tilde{\lambda}_{e,t} = \underbrace{\alpha'_{e,c} z_t + \beta'_{e,c} y_t}_{\text{credit intensity } \tilde{\lambda}_{e,c,t}} + \underbrace{\alpha'_{e,\ell} z_t + \beta'_{e,\ell} y_t}_{\text{liquidity intensity } \tilde{\lambda}_{e,\ell,t}}$$

- ▶ *In both papers, identification of credit/liquidity components based on:*

KfW-Bund spreads = liquidity spreads

(KfW bonds fully guaranteed against default by federal government).

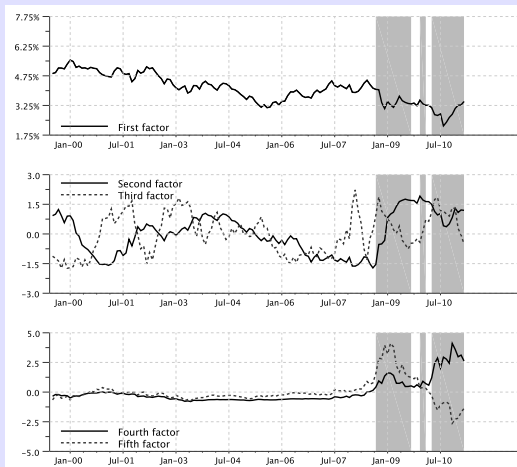
Example II.1 (Euro area sovereign credit spreads (cont'd))

- ▶ In MR2011, y_t is 5-dimensional vector of observed factors:
 - $(y_{1,t}, y_{2,t}, y_{3,t})' = (\text{level}, \text{slope}, \text{curvature})'$ of the German (Bund) yield curve.
 - $(y_{4,t}, y_{5,t})' = \text{PC1 and PC2 of 10-year spreads against Germany (France, Italy, Netherlands and Spain).}$
- ▶ 2 regimes, one is a “crisis regime”.
- ▶ Use of surveys in the estimation (see Slide 64).
- ▶ 2-step estimation approach (estimated factors shown on Figure 18):
 1. Historical dynamics of w_t estimated by Kitagawa-Hamilton (KH) filter (Appendix I.A).
 2. z_t fixed (based on smoothed estimates from KH).
Non-linear-least-squares techniques, taking into account the internal-consistency issue. More precisely, we estimate θ , the vector of parameters defining $\mathbb{Q}$ dynamics, by:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \left\{ \sum_{e,t,h} (R_{e,t,h}^{\text{obs}} - R_{e,t,h}(\theta))^2 \right\} + \chi \left\{ \sum_{t,i} (y_{i,t}^{\text{obs}} - y_{i,t}(\theta))^2 \right\},$$

where χ penalizes inconsistency between observed and model-implied observable factors y_t .

Figure 18: *Estimated factors*



Source: *Monfort and Renne (2011)*.

- ▶ In Monfort and Renne (2014), we have

$$\tilde{\lambda}_{e,t} = \underbrace{y_{e,t}}_{=\tilde{\lambda}_{e,c,t}} + \underbrace{(1 - \theta_\ell^{(e)})y_{E+1,t}}_{=\tilde{\lambda}_{e,\ell,t}}$$

hence y_t of dimension $E + 1 = 12$ (11 countries are involved).

- ▶ 9 regimes: $z_t = z_{c,t} \otimes z_{\ell,t}$, where $z_{i,t}$ ($i \in \{c, \ell\}$) is 3-dimensional.
 - $z_{i,t} = [1, 0, 0]'$: low stress,
 - $z_{i,t} = [0, 1, 0]'$: medium stress,
 - $z_{i,t} = [0, 0, 1]'$: high stress.
- ▶ $z_{c,t}$ and $z_{\ell,t}$ are not independent: for instance, proba. to go from low-credit-stress regime to medium/high-credit-stress regimes depends on the liquidity-stress regime.
- ▶ For each intensity $y_{i,t}$ (simulation on next slide):

$$y_{i,t} = \mu'_i z_t + \phi y_{i,t-1} + \sigma_i \varepsilon_{i,t},$$

with $\varepsilon_t \sim i.i.d. \mathcal{N}(0, \Sigma)$, Σ diagonal matrix and $\mu_{j,1} < \mu_{j,2} < \mu_{j,3}$.

- ▶ z_t is the only source of correlation across the $y_{i,t}$'s.
- ▶ SDF exponential affine in $(z_t', y_t')'$, same type of dynamics under $\mathbb{Q}$.

Example II.1 (Euro area sovereign credit spreads (cont'd))

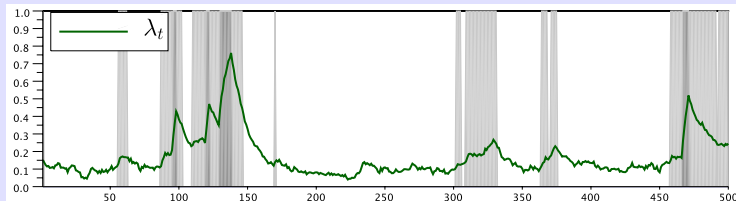
Generic process followed by the intensities (the $\lambda_{c,t}^{(n)}$'s and $\lambda_{\ell,t}$):

$$\lambda_t = \mu' z_t + \rho \lambda_{t-1} + \sigma \varepsilon_t$$

Simulated example:

with $\mu = [0.01 \quad 0.03 \quad 0.10]$, $P = \begin{bmatrix} 0.98 & 0.02 & 0 \\ 0.05 & 0.90 & 0.05 \\ 0 & 0.30 & 0.70 \end{bmatrix}$. and

$$\varepsilon_t \sim i.i.d. \mathcal{N}(0, 0.01^2)$$



Light grey: crisis regime ($z_t = [0, 1, 0]'$), Dark grey: severe crisis regime ($z_t = [0, 0, 1]'$)

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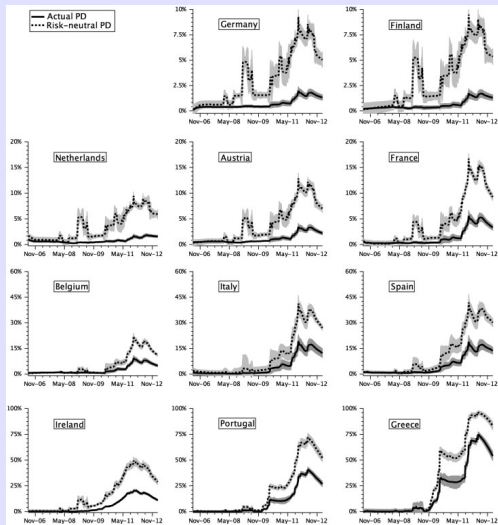
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Figure 19: 5-year Probabilities of Default



Source: Monfort and Renne (2014).

Example II.2 (IBOR-OIS spreads)

- ▶ *Dubecq et al. (2016)* decompose the EURIBOR-OIS spreads into a credit and a liquidity component. (OIS: see Def. II.1).
- ▶ N banks of EURIBOR panel (homogeneity assumption).
- ▶ EURIBOR rate = rate of an unsecured loan. From the point of view of bank j , lending to bank e ($e, j \in \{1, \dots, N\}$):

$$R_{t,h}^{IBOR} = -\frac{1}{h} \log \mathbb{E}_t^{\mathbb{Q}} \left\{ \exp \left[-\sum_{i=1}^h (r_{t+i-1} + \tilde{\lambda}_{e,c,t+i} + \tilde{\lambda}_{j,\ell,t+i}) \right] \right\}. \quad (\text{II.16})$$

- ▶ Overnight-Index Swap (OIS) rate:

$$\Rightarrow R_{t,h}^{OIS} = -\frac{1}{h} \log \mathbb{E}_t^{\mathbb{Q}} \left\{ \exp \left(-\sum_{i=1}^h r_{t+i-1} \right) \right\}. \quad (\text{II.17})$$

- ▶ r_t assumed to be independent from $\tilde{\lambda}'s$ + homogeneity assumption $\Rightarrow$

$$R_{t,h}^{IBOR} - R_{t,h}^{OIS} = -\frac{1}{h} \log \mathbb{E}_t^{\mathbb{Q}} \left\{ \exp \left[-\sum_{i=1}^h (\tilde{\lambda}_{c,t+i} + \tilde{\lambda}_{\ell,t+i}) \right] \right\}.$$

Example II.2 (IBOR-OIS spreads (cont'd))

- Quadratic specification for $\tilde{\lambda}_{c,t}$ and $\tilde{\lambda}_{\ell,t}$:

$$\tilde{\lambda}_{c,t} = x_{c,t}^2 \quad \text{and} \quad \tilde{\lambda}_{\ell,t} = x_{\ell,t}^2,$$

where $X_t = (x_{c,t}, x_{\ell,t})'$ follows a Gaussian VAR(1) process.

⇒ Quadratic Gaussian framework (see previous Lecture).

- X_t is latent. Estimation by Quadratic Kalman Filter (*Monfort, Renne and Roussellet, 2015*). Measurement equations include equations stating that observed proxies of credit and liquidity risks relate to quadratic functions of $x_{c,t}$ and $x_{\ell,t}$, respectively.

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Example II.2 (IBOR-OIS spreads (cont'd))

Figure 20: *Decomposition of 6M IBOR-OIS spreads*

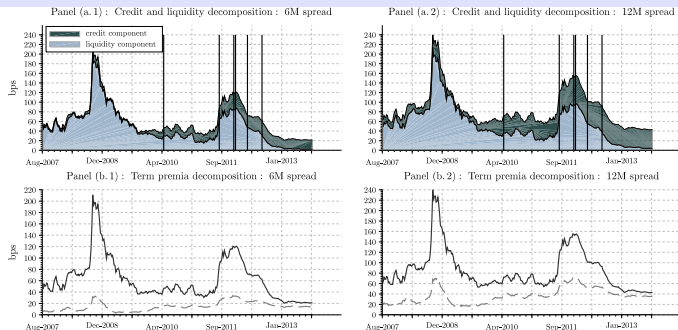


Fig. 4. 6M EURIBOR-OIS spreads decomposition. Notes: Date ranges from August 31 2007 to September 13 2013. Units are in basis points. Panels (a) represent the stacked components of the spread: light gray component is the liquidity component and the dark gray corresponds to the credit component. Panels (b) represent the modeled spread (black) and its term premia (gray dashed). The black vertical axes stand from left to right for: SMP program announcements (first two axes), VLTRO announcement and allotments (next three axes), and Mario Draghi's London speech (last axis).

Source: *Dubecq, Monfort, Renne and Roussellet (2016)*.

II.5 – RELAXING THE CLASSICAL ASSUMPTIONS

- ▶ Formula (II.12) and its tractability is a key feature of several models (among many others: Duffie and Singleton, 1999, Pan and Singleton, 2008, Monfort and Renne, 2014).
 - ▶ This formula is valid under Assumptions II.2 (non-systemic entities = “no-jump condition”), II.3 (unpriced credit risk vents) and II.5 (no contagion) $\equiv$ **classical credit risk models**.
 - ▶ We lose the high degree of tractability of Formula (II.12) when some of the previous assumptions are relaxed (see next slide).
- If $f(y_t | w_{t-1}) = f(y_t | y_{t-1}, d_{1,t-1})$ (entity $e = 1$ is “**systemic**”) then:
- ▶ The pseudo-price $\tilde{B}_e(t, h)$ now depends on y_t and $d_{1,t}$;
 - ▶ Eq. (II.5) is valid for entity $e = 2$ only;
 - ▶ The computation of $\tilde{B}_2(t, h)$ is not straightforward: $\{y_t\}$ is not autonomous and the autonomous process $\{y_t, d_{1,t}\}$ is not affine.

- If there is a contagion effect from $e = 1$ towards $e = 2$, i.e. if

$$\begin{aligned} p(d_t | y_t, w_{t-1}) &= p_1(d_{1,t} | y_t, y_{t-1}, d_{1,t-1}) \\ &\quad p_2(d_{2,t} | y_t, y_{t-1}, d_{1,t}, d_{1,t-1}, d_{2,t-1}) \quad \text{then:} \end{aligned}$$

- ▶ Eq. (II.5) is still valid for both entities;
 - ▶ In an affine framework, the computation of $\tilde{B}_1(t, h)$ (and $B_1(t, h)$) is straightforward; but formulas for $\tilde{B}_2(t, h)$ and $B_2(t, h)$ are not explicit anymore even if $\tilde{\lambda}_{2,t}$ is affine in $(y_t, y_{t-1}, d_{1,t}, d_{1,t-1})$.
- If the default event of the first entity only ($d_{1,t}$) is a source of risk that is priced: $M_{t-1,t}(w_t, w_{t-1}) = M_{t-1,t}(y_t, y_{t-1}, d_{1,t}, d_{1,t-1})$, then
- ▶ $\lambda_{1,t}^{\mathbb{Q}} \neq \lambda_{1,t}$ and $\lambda_{2,t}^{\mathbb{Q}} = \lambda_{2,t}$;
 - ▶ the exogeneity of $\{y_t\}$ is *no longer preserved* under $\mathbb{Q}$;
 - ▶ Eq. (II.5) is no longer valid for entity $e = 1$. It remains valid for $e = 2$ but the computation of $\tilde{B}_2(t, h)$ is not straightforward: even if y_t is $\mathbb{P}$ -autonomous, this is not true under $\mathbb{Q}$.

II.6 – GENERAL AFFINE CREDIT-RISK FRAMEWORK

- ▶ Exploiting VARG processes, Monfort, Pegoraro, Renne and Roussellet (2020) propose a general affine credit-risk pricing model jointly allowing for:
 - (i) **systemic entities** (breaking down the no-jump condition; see Collin-Dufresne, Goldstein and Hugonnier (2004));
 - (ii) **contagion effects between entities** (economic/financial linkages; see Aït-Sahalia, Laeven and Pelizzon (2014));
 - (iii) **pricing of credit events** (credit spread puzzle; see Gourieroux, Monfort and Renne (2014));
 - (iv) **and stochastic recovery rates (RR)** (Altman, Resti and Sironi (2005)).
- ▶ This general framework is presented in the following slides.
- ▶ State vector w_t of the form $[y_t', \delta_t']'$, where δ_t is a E -dimensional vector of **credit-event variables** (see Assumption II.6 on next slide).

Assumption II.6

The default date $\tau^{(e)}$ (say) of any entity e is defined as:

$$\tau^{(e)} = \inf \left\{ t > 0 : \delta_t^{(e)} > 0 \right\}, \quad (\text{II.18})$$

where $\delta_t^{(e)}$ is a non-negative variable called **credit-event variable**. The default indicator function can be equivalently written as $d_t^{(e)} = \mathbb{1}_{\{\tau^{(e)} \leq t\}}$ or $d_t^{(e)} = 1 - \mathbb{1}_{\{\underline{\delta}_t^{(e)'} \mathbf{1} = 0\}}$, with $\underline{\delta}_t^{(e)} = (\delta_t^{(e)}, \dots, \delta_1^{(e)})$ and where $\mathbf{1} = (1, \dots, 1)'$ with conformable dimension.

Assumption II.7

The stochastic process $\{w_t\}$ is affine under the historical probability measure $\mathbb{P}$. The historical Laplace transform of w_t , conditional $\underline{w}_{t-1}$, is denoted by:

$$\varphi_{w,t-1}^{\mathbb{P}}(u_w) = \mathbb{E} \left[\exp(u_w' w_t) \mid \underline{w}_{t-1} \right] = \exp [a_w(u_w)' w_{t-1} + b_w(u_w)], \quad (\text{II.19})$$

with $u_w = (u_y', u_\delta')'$.

Example II.3 ($w_t = (y_t', \delta_t')' =$ Vector Autoregressive Gamma process)

VARG (positive affine) process (see previous Lectures):

$$y_t \mid \underline{w_{t-1}} \stackrel{\mathbb{P}}{\sim} \bigotimes_{j=1}^{N_y} \gamma_{\nu_j^{(y)}} \left(\alpha_j^{(y)} + \overbrace{\beta_{j,y}^{(y)} y_{t-1}}^{\text{factors}} + \underbrace{\beta_{j,\delta}^{(y)} \delta_{t-1}}_{\text{credit-event variables}}, \mu_j^{(y)} \right)$$

kills no-jump cond.

$$\delta_t \mid y_t, \underline{w_{t-1}} \stackrel{\mathbb{P}}{\sim} \bigotimes_{e=1}^E \gamma_0 \left(\alpha_e^{(\delta)} + \beta_{e,y}^{(\delta)} y_t + \underbrace{\beta_{e,\delta}^{(\delta)} \delta_{t-1}}_{\text{contagion}}, \mu_e^{(\delta)} \right),$$

where $\delta_t = (\delta_t^{(1)}, \dots, \delta_t^{(E)})'$ and $\gamma_\nu(\lambda, \mu)$: non-central Gamma distribution.

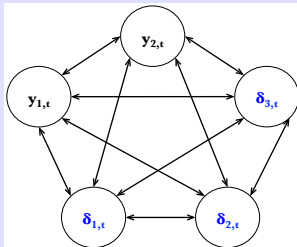
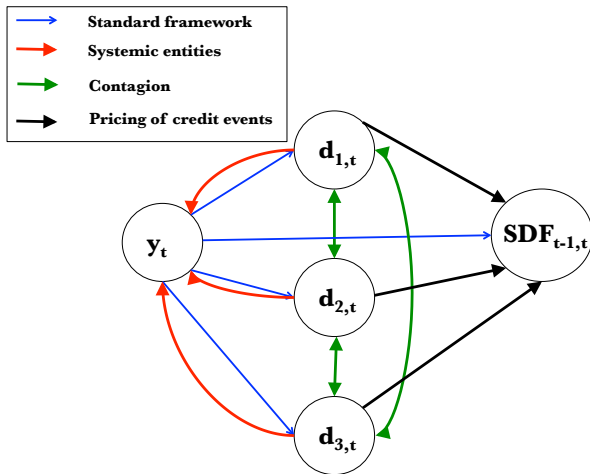


Figure 21: Causality scheme in the general credit-risk framework



(To be compared to Figure 16)

Assumption II.9 (General recovery payment)

Recovery Payment at date $t + i = \tau^{(e)}$ of a defaultable ZCB:

$$RR_{t+i}^{(e)} \times \mathcal{V}_{t+i, h-i}^{(e)} \quad (\text{II.24})$$

where the *Recovery Rate* $RR_{t+i}^{(e)}$ is

$$RR_{t+i}^{(e)} = \exp \left(-a_e - a'_{w,e} w_{t+i} \right), \quad (\text{II.25})$$

and where $\mathcal{V}_{t+i, h-i}^{(e)}$ denotes the *Recovery Value (Exposure-at-Default)* at $t + i$.

⇒ The general framework allows for flexible specifications of the recovery payment:

- (i) Fraction of “pre-default value” of the claim (Recovery of Market Value, or RMV) [see Prop. II.4],
 - (ii) Fraction of par (Recovery of Face Value, or RFV) [see Appendix II.C],
 - (iii) Fraction of a no-default version of the same claim (Recovery of Treasury, or RT) [see MPRR, 2018, Online Appendix A.5].
- Recovery payment = fraction of the (risk-free) present value of the principal.

Prop. II.4 (General pricing under the RMV convention)

If the recovery value at date $t + i$ (defined in II.9) is of the form:

$$\mathcal{V}_{t+i, h-i}^{(e)} = \mathbb{E}^{\mathbb{Q}} \left\{ \exp \left[- \sum_{j=i}^{h-1} \left(r_{t+j} + \delta_{t+j+1}^{(e)} \right) \right] \middle| \underline{w}_{t+i} \right\}, \quad (\text{II.26})$$

with $RR_t^{(e)} = \exp(-\delta_t^{(e)})$, and under Assumptions II.6 to II.8, the price $B_e(t, h)$ at date $t < \tau^{(e)}$ is given by:

$$B_e(t, h) = \mathcal{V}_{t, h}^{(e)}, \quad (\text{II.27})$$

where

$$\mathcal{V}_{t, h}^{(e)} = \exp \left[(\mathcal{A}_h - \xi_1)' w_t + (\mathcal{B}_h - h \xi_0) \right]. \quad (\text{II.28})$$

where $\mathcal{A}_h$ and $\mathcal{B}_h$ are obtained recursively and where $u_1 = -\tilde{e}_\delta$ and, for $i > 1$, $u_i = -(\tilde{e}_\delta + \xi_1)$ where $\tilde{e}_\delta = (0', e'_\delta)'$ is a N -dimensional vector, and where e_δ is the e^{th} column of the (N_δ, N_δ) -dimensional identity matrix.

Proof. See MPRR, 2018, Online Appendix A.4.

- Eq. (II.27) is a key result. It reads:

$$B_e(t, h) = \mathbb{E}^{\mathbb{Q}} \left\{ \exp \left[- \left(\sum_{i=1}^h r_{t+i-1} + \delta_{t+i}^{(e)} \right) \right] \mid \underline{w}_t \right\}. \quad (\text{II.29})$$

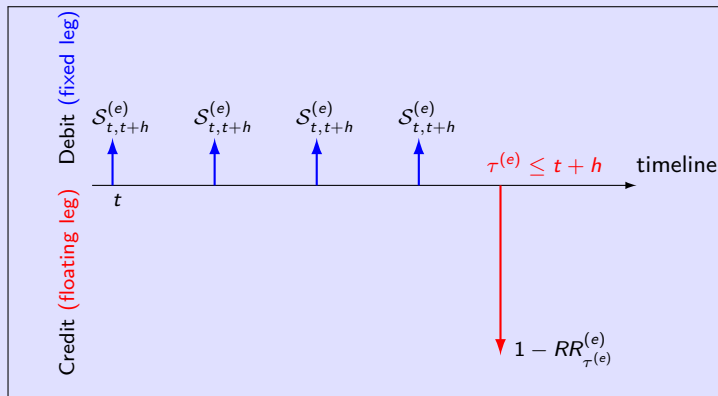
- It is reminiscent of Eq. (II.5), with λ (default intensity) replaced by δ (credit-event variable).
- However, contrary to the classical framework, in the present one:
- credit events are priced sources of risk,
 - the no-jump condition is relaxed,
 - contagion is allowed
 - and the recovery rate is stochastic.

⇒ Generalization of the Duffie and Singleton (1999) setting.

II.7 – CDS PRICING IN THE GENERAL FRAMEWORK

CDS payoffs, with payoffs in the domestic currency

Figure 22: At t , purchase of credit protection, maturity date $t + h$



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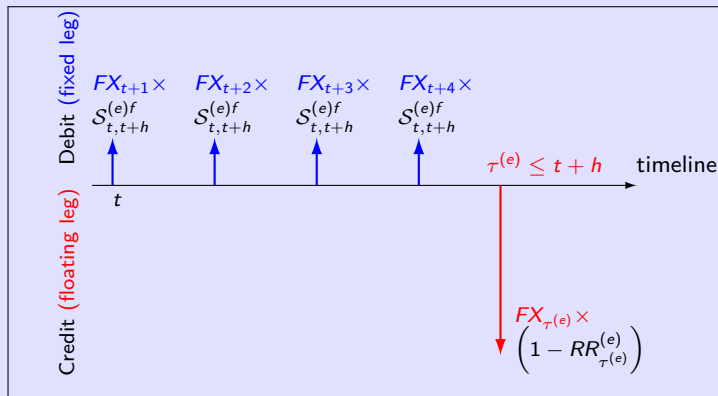
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II.7 – CDS PRICING IN THE GENERAL FRAMEWORK

CDS payoffs, with payoffs in the foreign currency

Figure 23: At t , purchase of credit protection, maturity date $t + h$



II.7 – CDS PRICING IN THE GENERAL FRAMEWORK

- ▶ The domestic-currency case is a special case of the multi-currency one.
- ▶ Pricing a CDS amounts to determining $S_{t,t+h}^{(e)}$ (**CDS spread**) for $t < \tau^{(e)}$.
- ▶ $S_{t,t+h}^{(e)}$ such that the date- t value of the **fixed leg**'s payoffs is equal to the date- t value of the **floating leg**'s payoffs.
- ▶ The payoffs of the assets issued by the reference entity are assumed to be expressed in the domestic currency.
- ▶ Notional and premium payments expressed in the currency of denomination.
- ▶ (Logarithmic) exchange rate: $s_t = \ln(FX_t)$.
Convention: increase in s_t = depreciation of the domestic currency.

II.7 – CDS PRICING IN THE GENERAL FRAMEWORK

- ▶ We consider a CDS whose notional is equal to one unit of the foreign currency (i.e. to $\exp(s_t)$ units of the domestic currency).

- ▶ **Fixed Leg:**

- If entity e has not defaulted at date $t + i$ ($\leq t + h$), the cash flow on this date, expressed in the domestic currency, is:

$$S_{t,t+h}^{(e)f} \exp(s_{t+i}).$$

- PV of the fixed-leg payments, expressed in the domestic currency:

$$S_{t,t+h}^{(e)f} \sum_{i=1}^h \mathbb{E}^{\mathbb{Q}} \left[\exp \left(s_{t+i} - \sum_{j=1}^i r_{t+j-1} \right) \mathbb{1}_{\{\delta_{t:t+i}^{(e)'} = 0\}} \mid \underline{w}_t \right],$$

► **Floating Leg:**

- Under the RFV convention, the protection seller will make a payment of $(1 - RR_{t+i}^{(e)}) \exp(s_{t+i})$ (*Loss-Given-Default*) at date $t + i$ in case of default over the time interval $]t + i - 1, t + i]$.
- PV of the floating leg expressed in the domestic currency:

$$\sum_{i=1}^h \mathbb{E}^{\mathbb{Q}} \left[\exp \left(s_{t+i} - \sum_{j=1}^i r_{t+j-1} \right) (1 - RR_{t+i}^{(e)}) \right. \\ \left. \left(\mathbb{1}_{\{\delta_{t:t+i-1}^{(e)'} \mathbf{1}=0\}} - \mathbb{1}_{\{\delta_{t:t+i}^{(e)'} \mathbf{1}=0\}} \right) \middle| \underline{w_t} \right],$$

- We assume $RR_t^{(e)} = \exp(-a_e - a'_{w,e} w_t)$ and $s_t = \chi + u'_{s,e} w_t$.

II.7 – CDS PRICING IN THE GENERAL FRAMEWORK

Prop. II.5 (Price of a multi-currency CDS)

$$\mathcal{S}_{t,t+h}^{(e)f} = \frac{\sum_{i=1}^h \Lambda_{(t,i)}^Q}{\sum_{i=1}^h \lim_{u \rightarrow -\infty} \Psi_{(t,i)}^Q(\chi, u\tilde{e}_\delta - \xi_1, u\tilde{e}_\delta + u_s)}, \quad (II.30)$$

where:

$$\begin{aligned} \Lambda_{(t,i)}^Q = \lim_{u \rightarrow -\infty} & \left[\Psi_{(t,i)}^Q(\chi, u\tilde{e}_\delta - \xi_1, u_s) - \Psi_{(t,i)}^Q(\chi, u\tilde{e}_\delta - \xi_1, u\tilde{e}_\delta + u_s) \right. \\ & - \Psi_{(t,i)}^Q(\chi - a_e, u\tilde{e}_\delta - \xi_1, u_s - a_w) \\ & \left. + \Psi_{(t,i)}^Q(\chi - a_e, u\tilde{e}_\delta - \xi_1, u\tilde{e}_\delta + u_s - a_w) \right], \end{aligned}$$

and where $\Psi_{(t,i)}^Q(\kappa, u_2, u_1)$ is given in Eq. (II.38).

Proof. See Online Appendix A.4 of MPRR (2018). ■

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II.8 – TOP-DOWN APPROACH

- ▶ Relaxing the assumptions underlying the classical framework can also be done in the context of “**top-down**” approaches.
- ▶ “Top-down” models focus on default counting (or loss) processes [e.g. [Azizpour, Giesecke, Baeho \(2011\)](#), [Giesecke, Goldberg, Ding \(2011\)](#)], contrary to the approach presented above (“bottom-up”) that consider default processes of individual firms as the model primitives [e.g. [Lando \(1998\)](#), [Duffie and Singleton \(1999\)](#), [Duffie and Gârleanu, 2001](#)].
- ▶ The “top-down” approach has been shown to satisfactorily capture the existence of default clustering [[Brigo, Pallavicini, Torresetti \(2007\)](#), [Errais, Giesecke and Goldberg, 2010](#)].
- ▶ Building on [Gouriéroux, Monfort and Renne \(2014\)](#), [Gouriéroux, Monfort, Mouabbi and Renne \(2018\)](#) propose an affine “Top-down” model consistent with: the presence of systemic entities, contagion, and the pricing of default events.

Example II.4 ("Disastrous Defaults")

- ▶ Model proposed by *Gouriéroux, Monfort, Mouabbi and Renne (2018)*.
- ▶ n_t^s : Number of systemic defaults on date t .
- ▶ N_t^s : number of systemic entities in default at date t ($N_t^s = n_t^s + N_{t-1}^s$).
- ▶ Conditional distribution of the number of systemic defaults:

$$n_{t+1}^s | \underline{x_{t+1}}, \underline{y_{t+1}}, \underline{N_t^s} \sim \text{Poisson}(\beta y_{t+1} + c n_t^s), \quad (\text{II.31})$$

where y_t is a nonnegative factor fluctuating around a low-frequency component x_t .

- ▶ $(y_t, x_t)'$ follows a VARG dynamics. This dynamics admits the following VAR representation:

$$\begin{cases} y_t - x_t &= \rho_y (y_{t-1} - x_{t-1}) + \sigma_{y,t} \varepsilon_{y,t} \\ x_t - \mu_x &= \rho_x (x_{t-1} - \mu_x) + \sigma_{x,t} \varepsilon_{x,t}, \end{cases} \quad (\text{II.32})$$

with $0 < \rho_y < \rho_x < 1$.

- ▶ If $c > 0$: Defaults on date t increases the conditional probability of having additional defaults on the next date $\Rightarrow$ *Systemic defaults are infectious [Davis and Lo, 2001], or contagious.*

Example II.4 ("Disastrous Defaults" (cont'd))

- $\Delta c_t = \log(C_t/C_{t-1})$: Log growth rate of per capita consumption:

$$\Delta c_t = \mu_{c,0} + \mu_{c,x}x_t + \mu_{c,y}y_t + \mu_{c,z}z_t. \quad (II.33)$$

where z_t depends on systemic defaults:

$$z_t | \underline{x}_t, \underline{y}_t, \underline{N}_t^s \sim \gamma_0(\xi_z n_{t-1}^s, \mu_z). \quad (II.34)$$

- γ_0 is the Gamma-Zero distribution featuring a point mass at zero.
 ⇒ The conditional probability that $z_t = 0$ is $\exp(-\xi_z n_{t-1}^s)$,
 $z_t = 0$ as long as there has been no systemic defaults in the
 previous period, which is rather frequent.
- If $\mu_{c,z} < 0$ and $|\mu_{c,z}|$ is large or
 if $\mu_{c,z} < 0$ and $|\mu_{c,z}|$ not so large but c (contamination) is large, then
 systemic defaults can give rise to "disastrous" decreases in C_t .
- Agents feature *Epstein-Zin preferences*, with a unit elasticity of
 intertemporal substitution (EIS). An admissible s.d.f. is exponential
 affine.
- ⇒ $w_t = (x_t, y_t, z_t, n_t^s)'$ is affine under $\mathbb{P}$ and $\mathbb{Q}$.

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Figure 24: Causality schemes

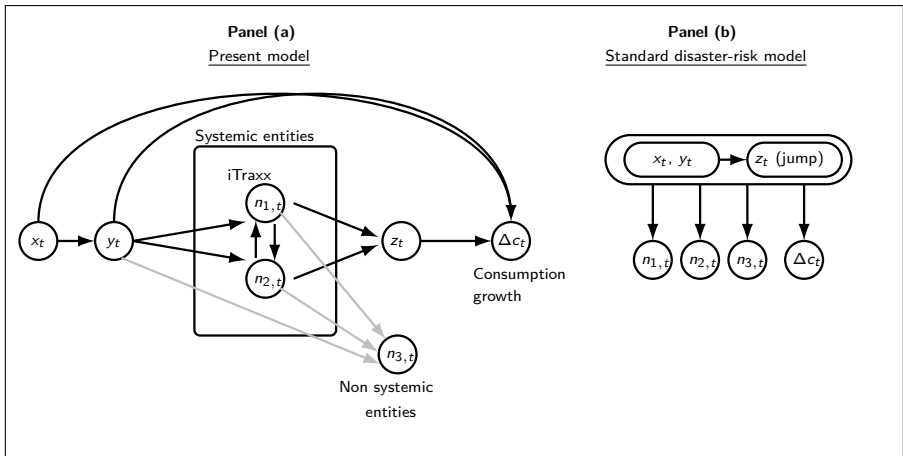


Figure 25: Synthetic Collateralised Debt Obligations (CDOs)

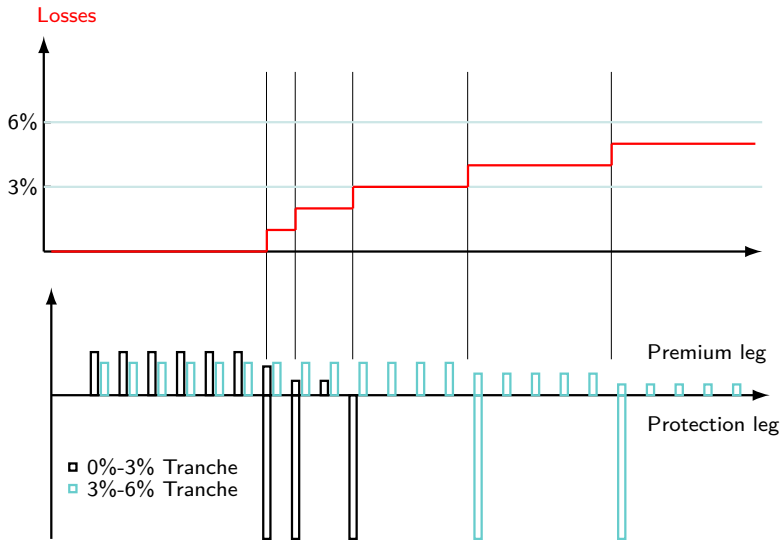
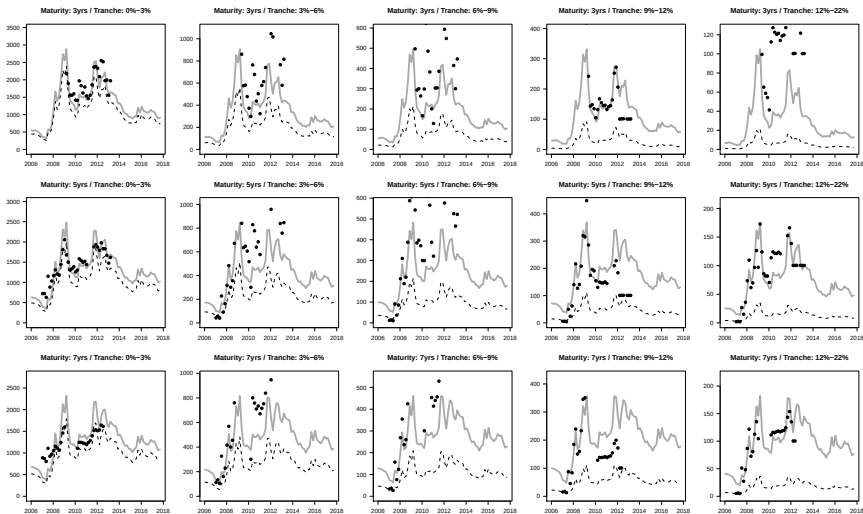


Figure 26: Fit of iTraxx tranches (grey:fitted, dashed: without risk premiums))



Appendix II.A (Structural interpretation of θ_ℓ (abstracting from credit risk))

- ▶ Each offer is a random fraction ω_i ($i \in \{1, \dots, \kappa\}$) of $\tilde{B}_e(t, h)$ (see Eq. II.15). The ω_i 's are uniformly distributed in $[0, 1]$.
- ▶ At $t + \varepsilon$, the bond is sold to the trader that has offered the highest price.
- ▶ Formally, when $\ell_t = 1$, the selling price is:

$$\left(\max_{i \in \{1, \dots, \kappa\}} \omega_i \right) \tilde{P}_{t, h}^{(e)}.$$

- ▶ Conditional on κ , the expectation of $\max_i(\omega_i)$ is $\kappa/(\kappa + 1)$. (Because the ω_i are i.i.d., the c.d.f. of $\max(\omega_1, \dots, \omega_\kappa)$ is $x \mapsto F(x)^\kappa$, where F is the c.d.f. of ω_i .)
- ▶ $\theta_\ell^{(e)}$, which is the unconditional expectation of $\max_i(\omega_i)$, is given by:

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{e^{-\varepsilon\gamma^{(e)}}}{k+1} \frac{(\varepsilon\gamma^{(e)})^k}{k!} &= \left[\left(1 - e^{-\varepsilon\gamma^{(e)}}\right) \frac{\varepsilon\gamma^{(e)} - 1}{\varepsilon\gamma^{(e)}} + e^{-\varepsilon\gamma^{(e)}} \right] \\ &= g(\varepsilon\gamma^{(e)}) \quad (\text{say}). \end{aligned}$$

- ▶ Function g is increasing and valued in $[0, 1]$. In addition $\lim_{x \rightarrow \infty} g(x) \rightarrow 1$.

Relation (II.10) is true for $B(t+h, 0)$ since $\widetilde{B}(t+h, 0) = 1$ and $B(t+h, 0) = 1 - d_{t+h}$. Assuming that $B(t+i+1, h-i-1) = (1 - d_{t+i+1})\widetilde{B}(t+i+1, h-i-1)$ (which is valid for $i = h-1$), we get:

$$B(t+i, h-i) = (1-d_{t+i}) \mathbb{E}_{t+i}^{\mathbb{Q}} \left\{ \exp(-r_{t+i}) \left[(1-d_{t+i+1}) \widetilde{B}(t+i+1, h-i-1) + d_{t+i+1} \zeta \widetilde{B}(t+i+1, h-i-1) \right] \right\},$$

since, in case of no default at date $t + i$, the value of the bond at date $t + i + 1$ is either $\zeta B(t + i + 1, h - i - 1) = \zeta \tilde{B}(t + i + 1, h - i - 1)$ if default happens, and $B(t + i + 1, h - i - 1) = \tilde{B}(t + i + 1, h - i - 1)$ otherwise. Assumptions II.2 and II.3 imply that $\tilde{B}(t + i + 1, h - i - 1)$ does not depend on d_{t+i+1} and taking first the conditional expectation given w_{t+i} and y_{t+i+1} we obtain:

$$B(t+i, h-i) = (1-d_{t+i})\mathbb{E}_{t+i}^{\mathbb{Q}}\left\{\exp(-r_{t+i})\widetilde{B}(t+i+1, h-i-1)\times\right. \\ \left.\mathbb{E}^{\mathbb{Q}}[(1-d_{t+i+1})+\zeta_{t+i+1}|y_{t+i+1}, w_{t+i}]\right\} . \quad (\text{II.35})$$

Using the definition of $\tilde{\lambda}_t$ (Eq. II.8) we get:

$$\begin{aligned} B(t+i, h-i) &= (1-d_{t+i}) \mathbb{E}_{t+i}^{\mathbb{Q}} \left\{ \exp \left(-r_{t+i} - \widetilde{\lambda}_{t+i+1} \right) \widetilde{B}(t+i+1, h-i-1) \right\} \\ &= (1-d_{t+i}) \widetilde{B}(t+i, h-i). \end{aligned}$$

Therefore it is also true for $i = 0$.

